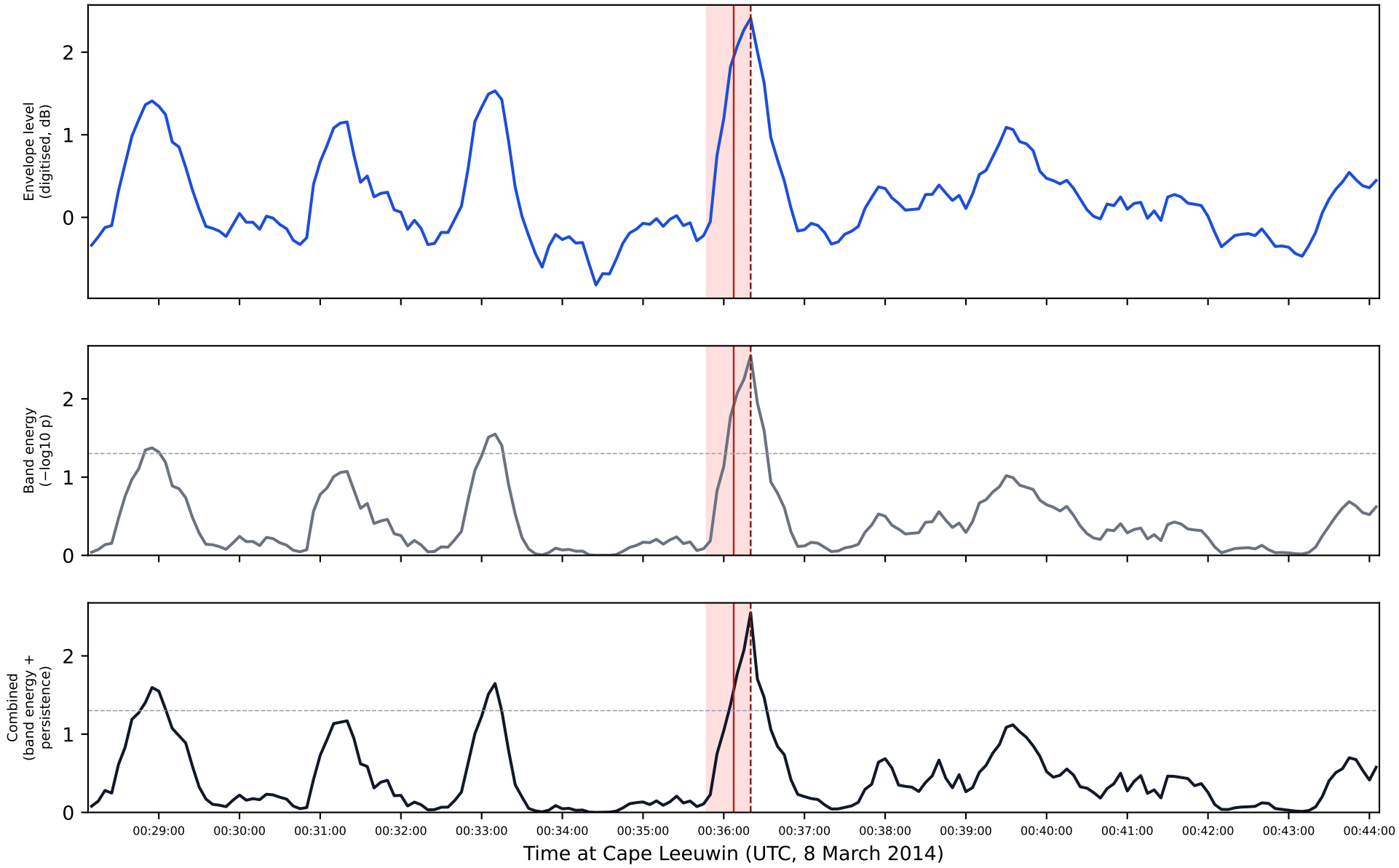
Ten strongest Cape Leeuwin and Diego Garcia South candidates with an implied source on the 7th arc

Rank	Station	Arrival	Bearing, published / corrected	Found by	Envelope p	Arc latitude (° S)	Implied impact	Steep dive	Time coincidence (label only)
1	Cape Leeuwin	00:36:07	none	ours	0.003	any (no bearing)	00:19:38–00:22:41	yes	Royer B, figure pick +0 s
2	Cape Leeuwin	00:47:04	343.2° / 338.7° (±2.0°)	Kadri and ours	0.006	15; 16 (outside 20-41)	00:19:38–00:27:03	yes	
3	Diego Garcia S.	01:10:15	none	ours	0.009	any (no bearing)	00:21:52–00:34:29	yes	
4	Cape Leeuwin	00:54:30	306.2° / 299.4° (±3.9°)	Kadri and ours	0.062	27	00:32:23–00:40:58	no	Royer 5 (uncertain), table pick -18 s
5	Cape Leeuwin	00:51:50	320.4° / 315.5° (±3.9°)	Kadri and ours	0.096	23; 24	00:27:47–00:37:50	no	Royer A, figure pick +7 s
6	Cape Leeuwin	00:49:58	260.4° / 246.8° (±2.1°)	Kadri	0.121	38; 39; 40	00:19:38–00:26:37	yes	
7	Cape Leeuwin	00:55:07	314.1° / 308.5° (±3.9°)	Kadri	0.237	25; 26	00:32:09–00:41:11	no	
8	Cape Leeuwin	00:45:36	312.0° / 306.1° (±3.9°)	Kadri	0.273	25; 26	00:23:02–00:31:40	yes	
9	Cape Leeuwin	00:40:34	303.3° / 296.0° (±3.9°)	Kadri	0.332	28	00:19:38–00:27:16	yes	
10	Cape Leeuwin	00:41:43	310.0° / 303.8° (±3.9°)	Kadri	0.344	26; 27	00:19:38–00:28:01	yes	

Ranking: envelope evidence at the station itself (band energy and persistence on the digitised Kadri (2024) Figure 9 envelope, smallest p within ±15 s; within-trace null). This is weak evidence: the p floor is about 1/355 and the 19 Kadri times are not corrected for multiple looks. Sources: Kadri Table 1; ours = combined-detector best looks on the envelopes and 4 Nov screening picks (merged within 30 s of a Kadri time, or 45 s of each other). Bearings: Kadri's published bearings corrected for the H01W array-position bias (calibration from 144 earthquakes, ruling B1; corrections -4.4 to -13.8°); tolerance ±1.645σ with σ = calibrated sector error and a 1.2° weak-signal term. Stronger candidates not shown: 00:39:53 (published 256.8°, corrected 243.0°; envelope p 0.093): bearing fits 40-43 S, but implied impact 23:57:02-00:12:19 is before 00:19:38; 00:49:41 (published 234.7°, corrected 221.7°; envelope p 0.121): corrected bearing meets no 7th-arc point; 00:38:29 (published 261.2°, corrected 247.7°; envelope p 0.214): bearing fits 38-43 S, but implied impact 23:55:06-00:15:19 is before 00:19:38; 00:39:02 (published 268.2°, corrected 255.3°; envelope p 0.251): bearing fits 36-40 S, but implied impact 00:05:12-00:19:30 is before 00:19:38. Implied impact: arc points 15-43° S within the bearing tolerance (no bearing: 20-41° S), sound speed: physical bounds per path (WOA23, 5 % detour) together with Kadri's 1.45 to 1.57 km/s, arrival span from the onset minus 20 s to the envelope peak, inside 00:19:38 to 00:55:00. Steep dive: impact 00:19:38 to 00:29:44 within 25 NM. Catalogue labels are associations below 0.95 probability (label only, not removed).

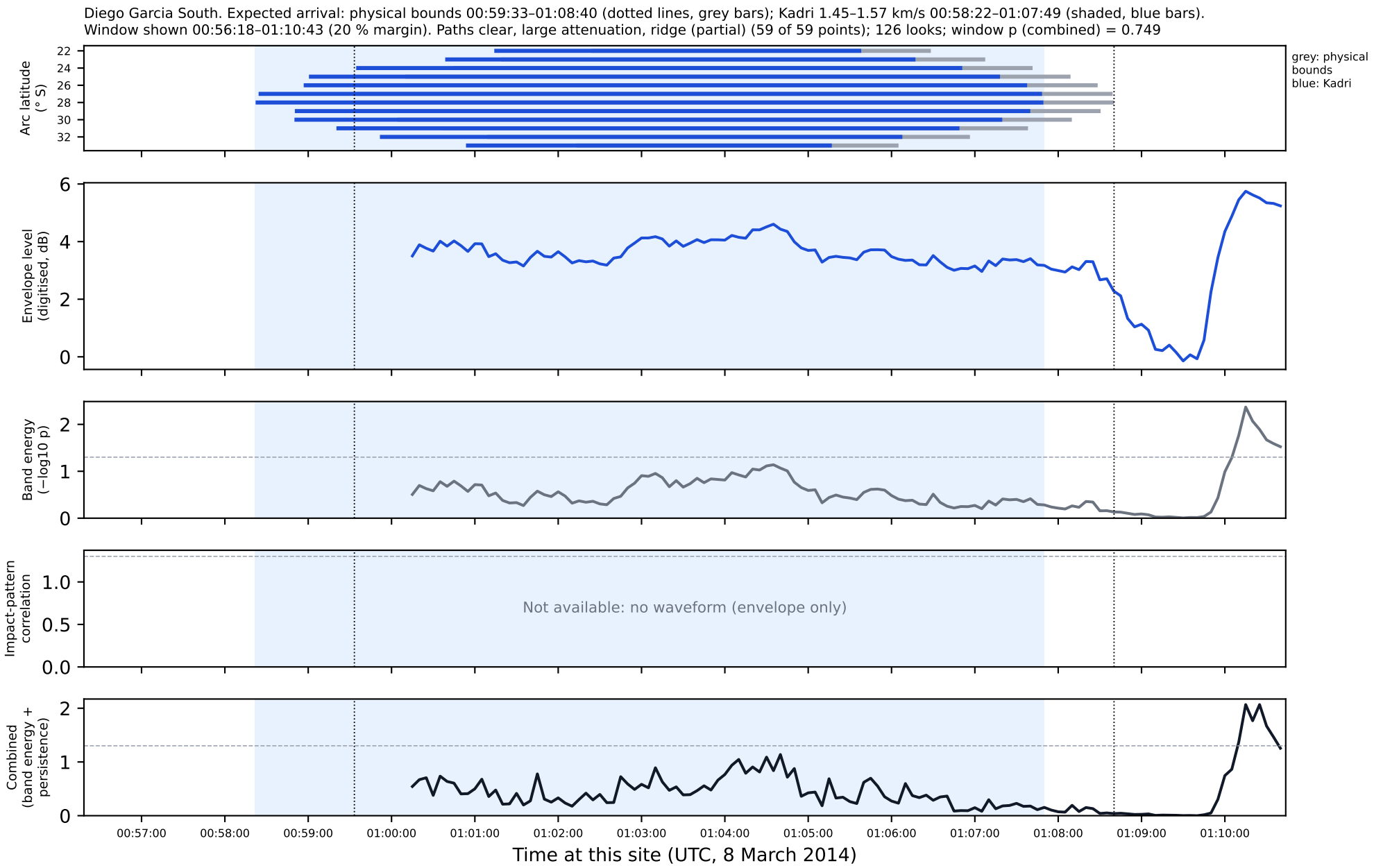
Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:22:41 UTC, arc latitude from timing only

Primary site, Cape Leeuwin. Solid red line: onset pick 00:36:07. Dashed line: envelope peak 00:36:20. Red band: arrival span used for timing, 00:35:47–00:36:20.
Envelope evidence p = 0.00282 (smallest within ±15 s of the onset; within-trace null).
Coincides in time with Royer B, figure pick +0 s; Royer B (uncertain), table pick -13 s (label only, not removed; Royer's event locations rest on his own unverified pick association).



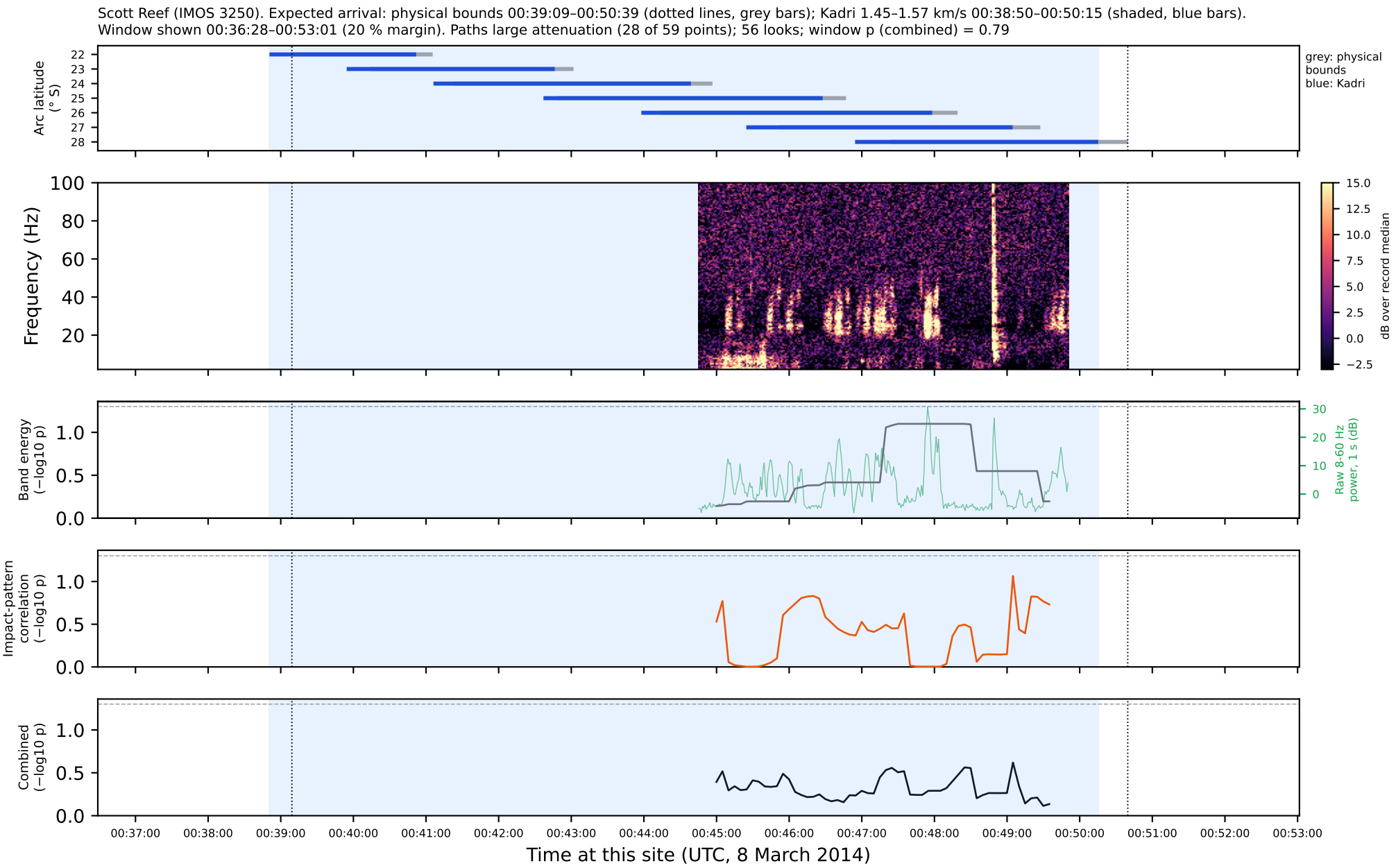
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:22:41 UTC, arc latitude from timing only



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

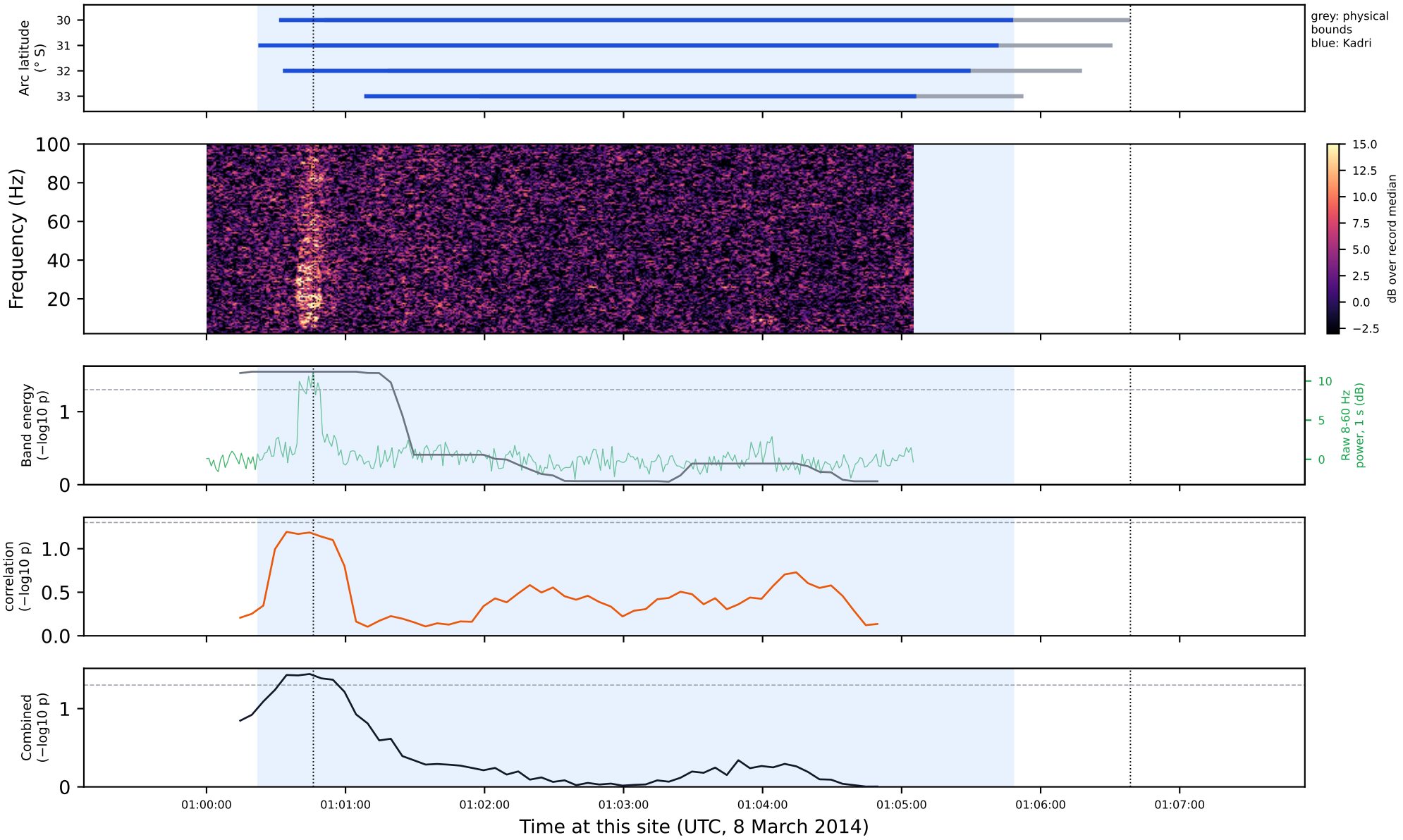
Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:22:41 UTC, arc latitude from timing only



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

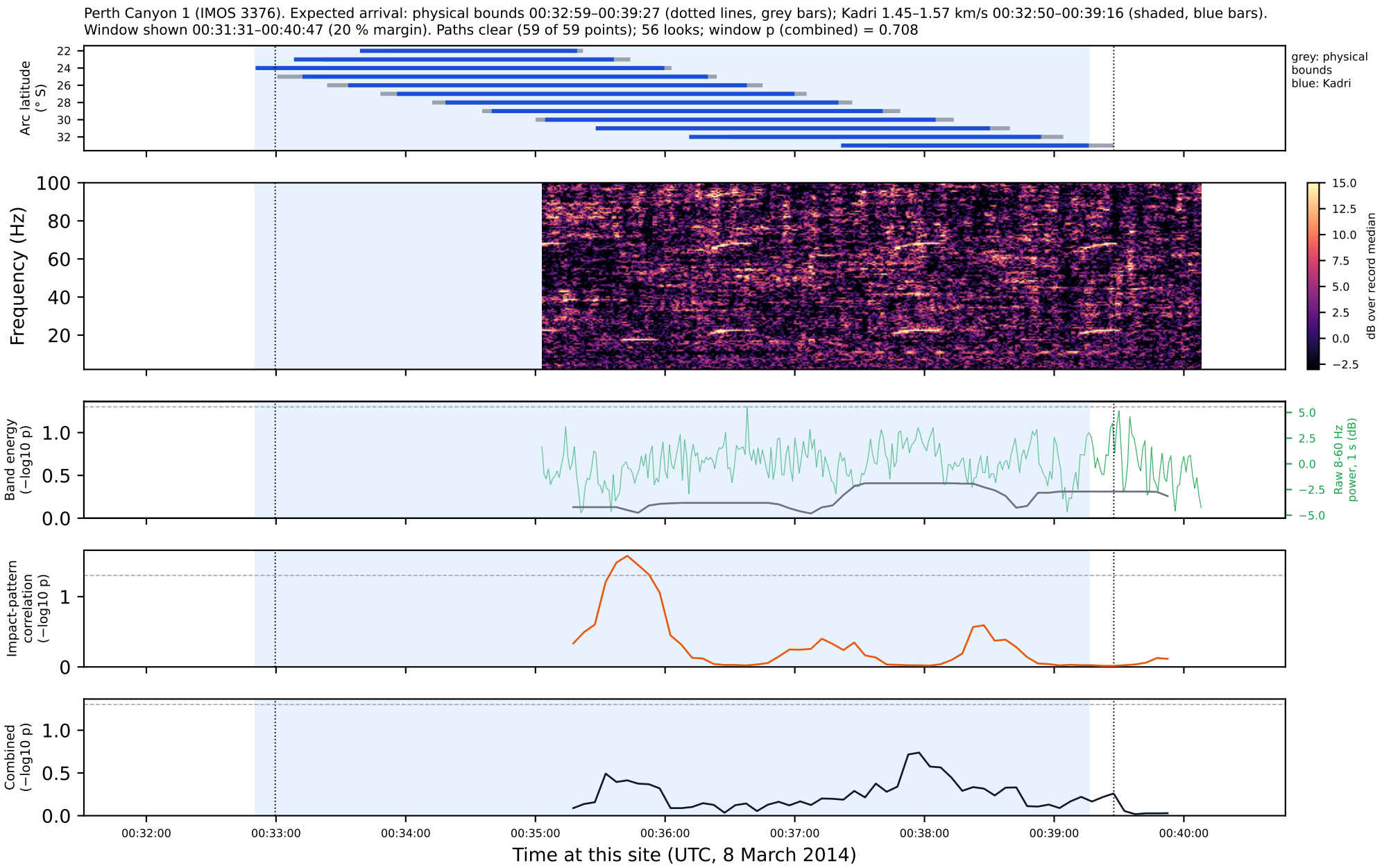
Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:22:41 UTC, arc latitude from timing only

Portland 1 (IMOS 3274). Expected arrival: physical bounds 01:00:46–01:06:38 (dotted lines, grey bars); Kadri 1.45–1.57 km/s 01:00:22–01:05:48 (shaded, blue bars).
Window shown 00:59:07–01:07:54 (20 % margin). Paths clear (17 of 59 points); 56 looks; window p (combined) = 0.248



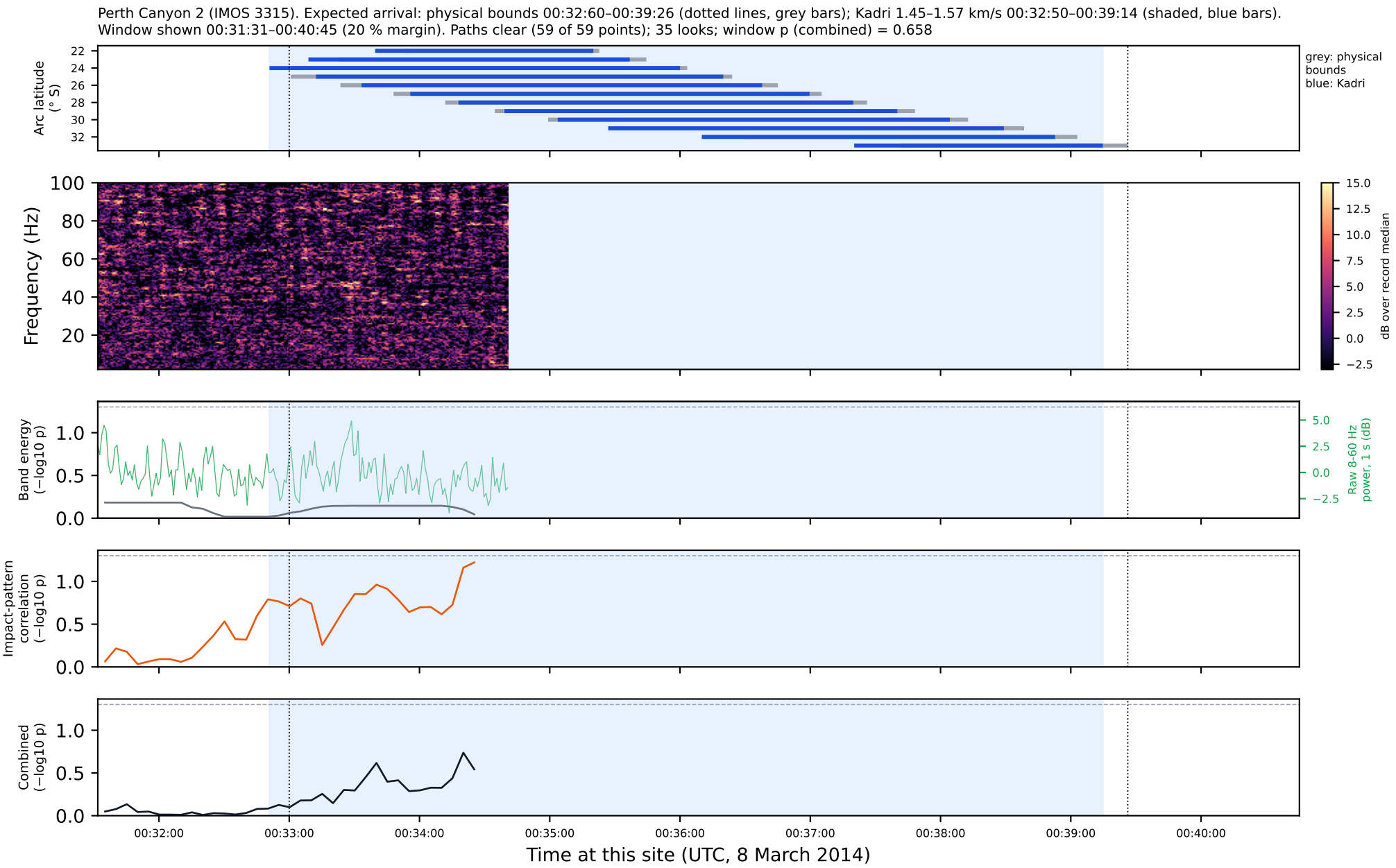
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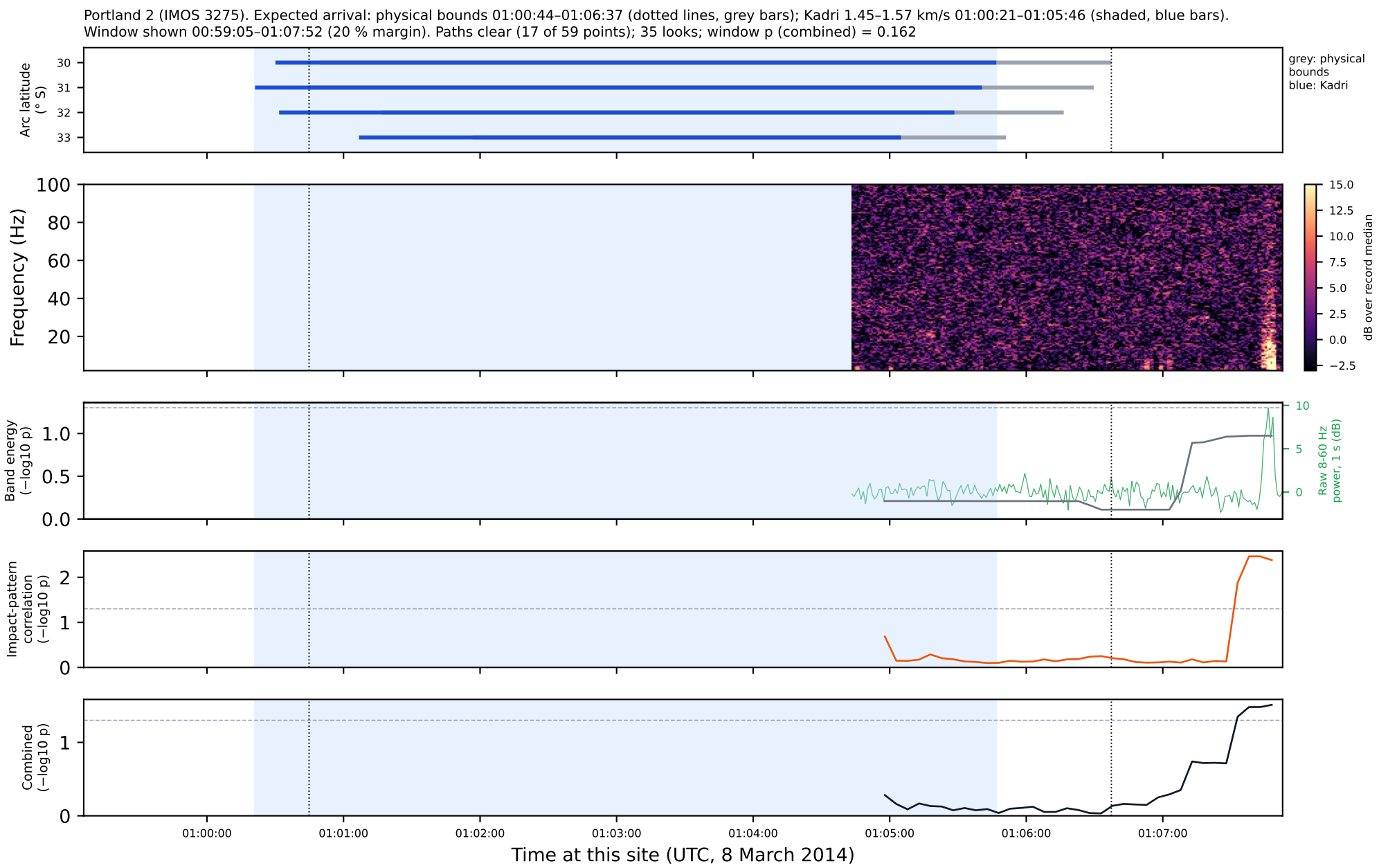
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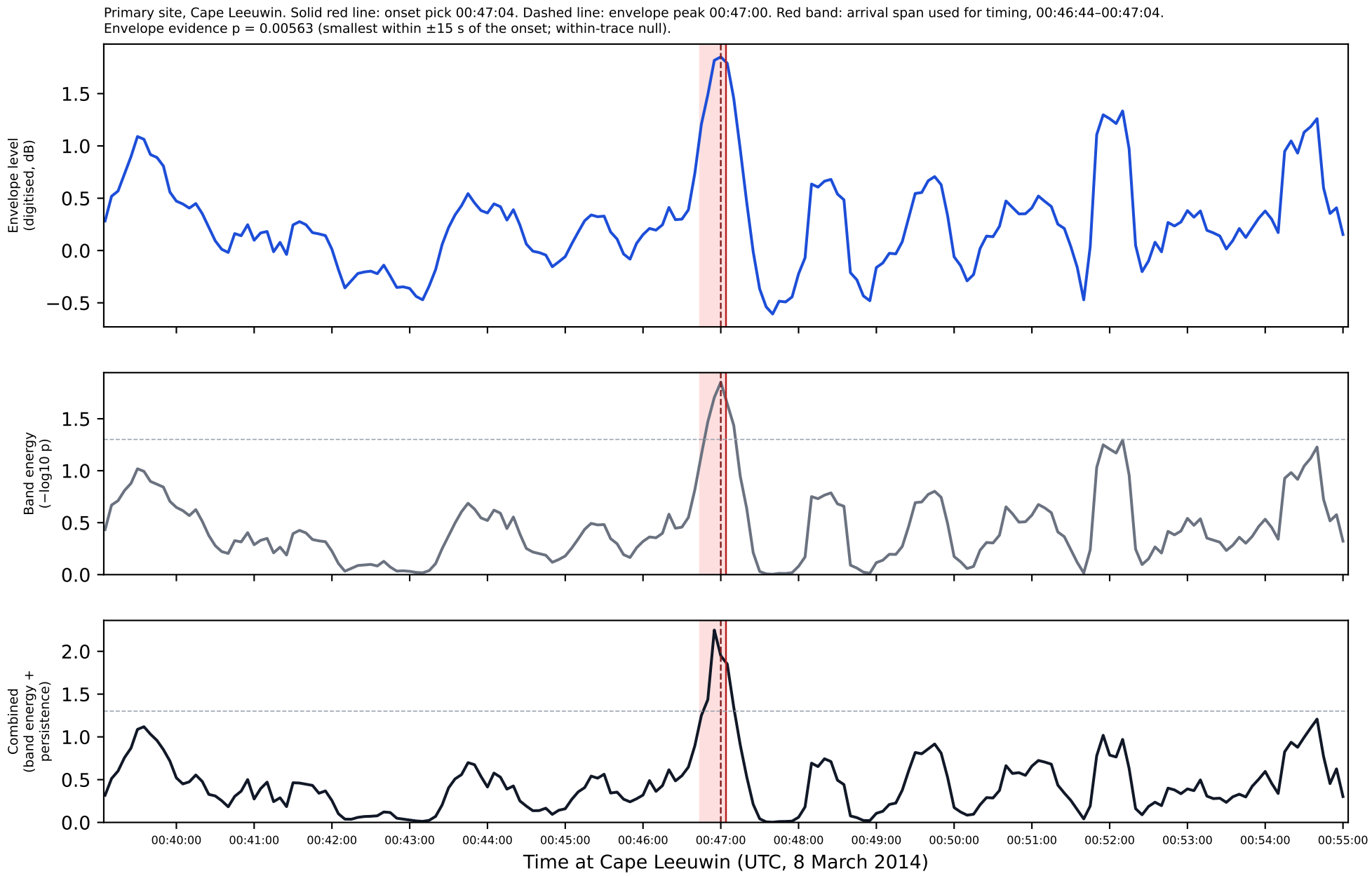
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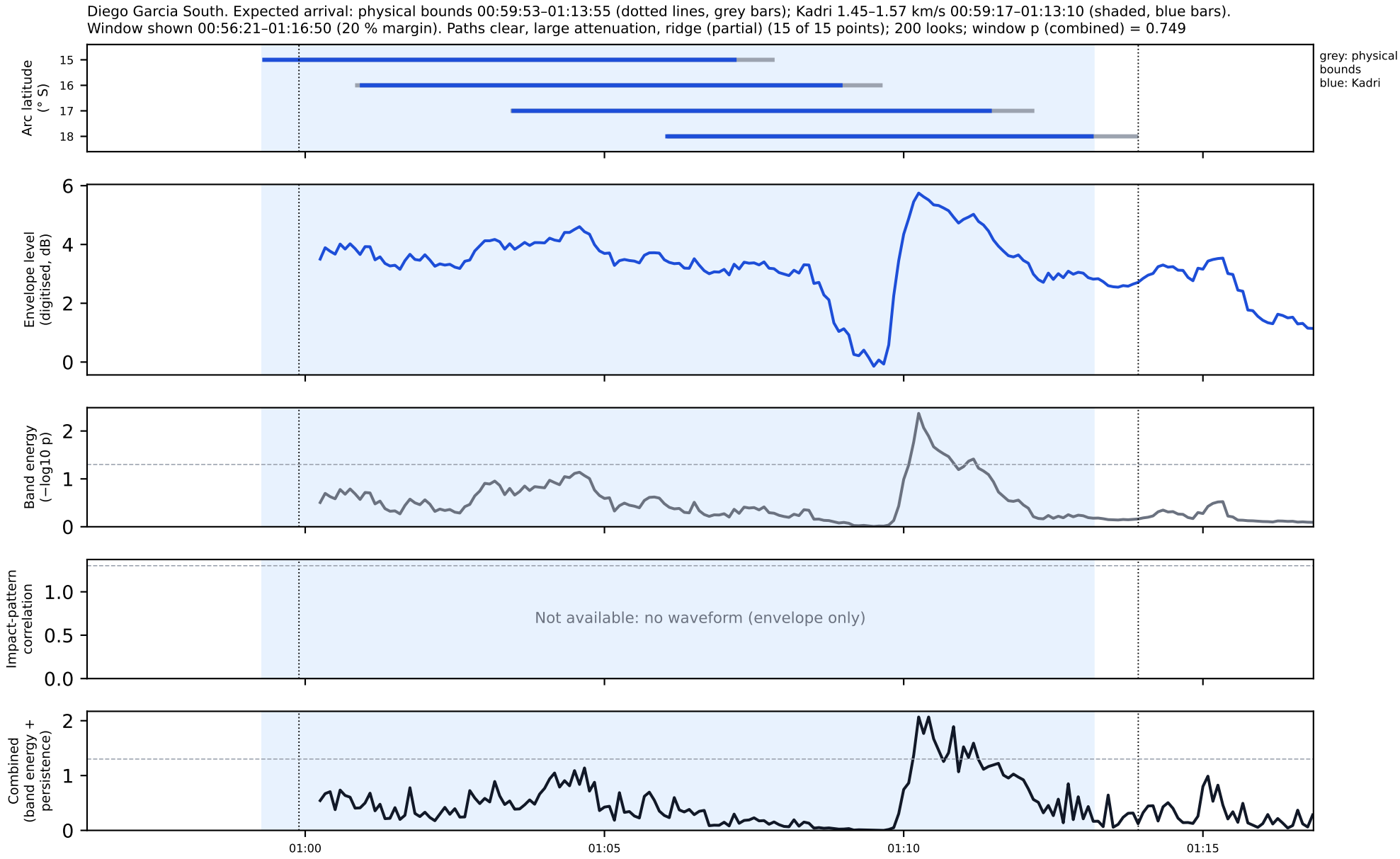
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Candidate 2: Cape Leeuwin 00:47:04 (Kadri and ours; bearing 338.7° corrected, 343.2° published). Implied impact 00:19:38–00:27:03 UTC, arc 15; 16° S



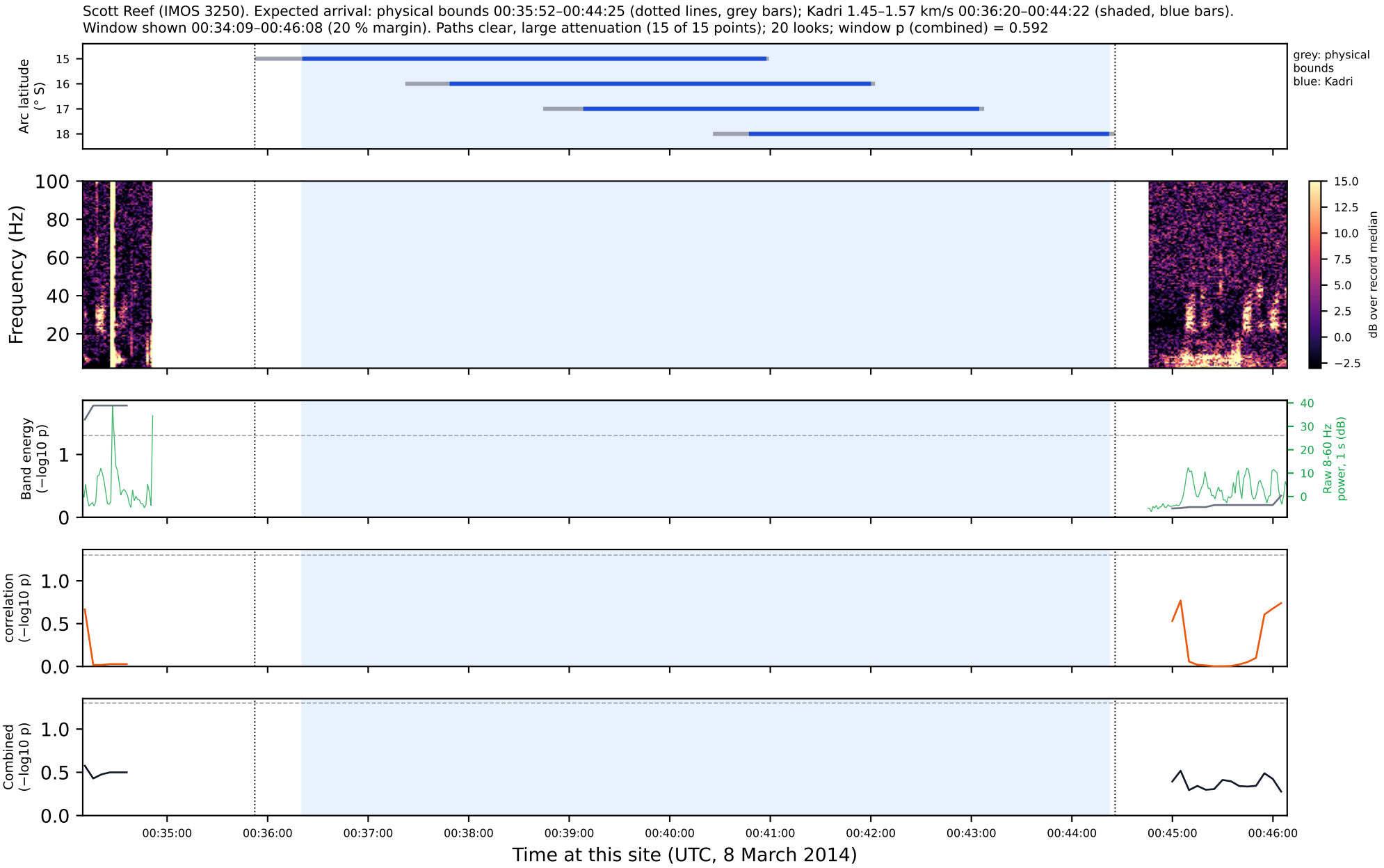
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

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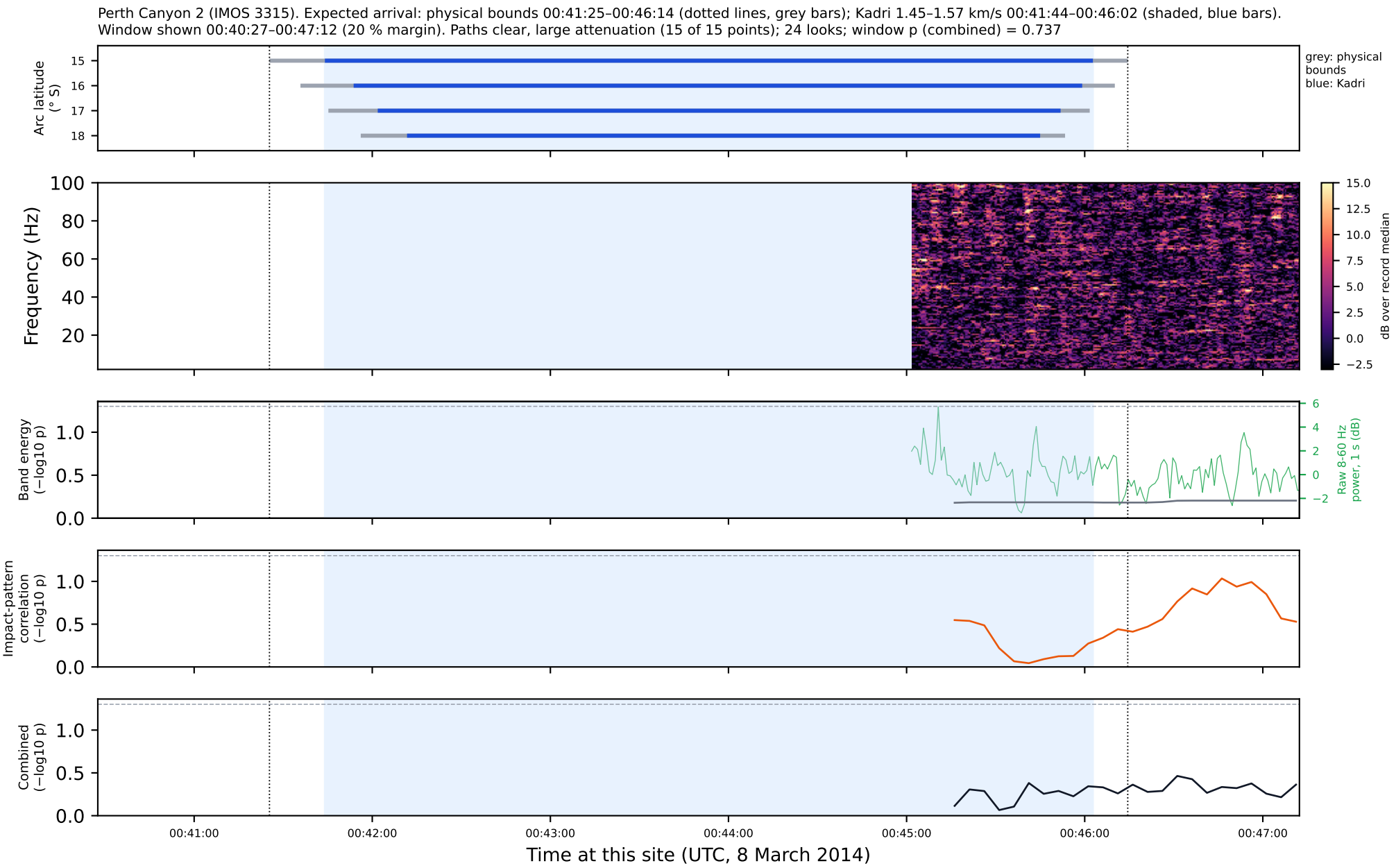
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Candidate 2: Cape Leeuwin 00:47:04 (Kadri and ours; bearing 338.7° corrected, 343.2° published). Implied impact 00:19:38–00:27:03 UTC, arc 15; 16° S



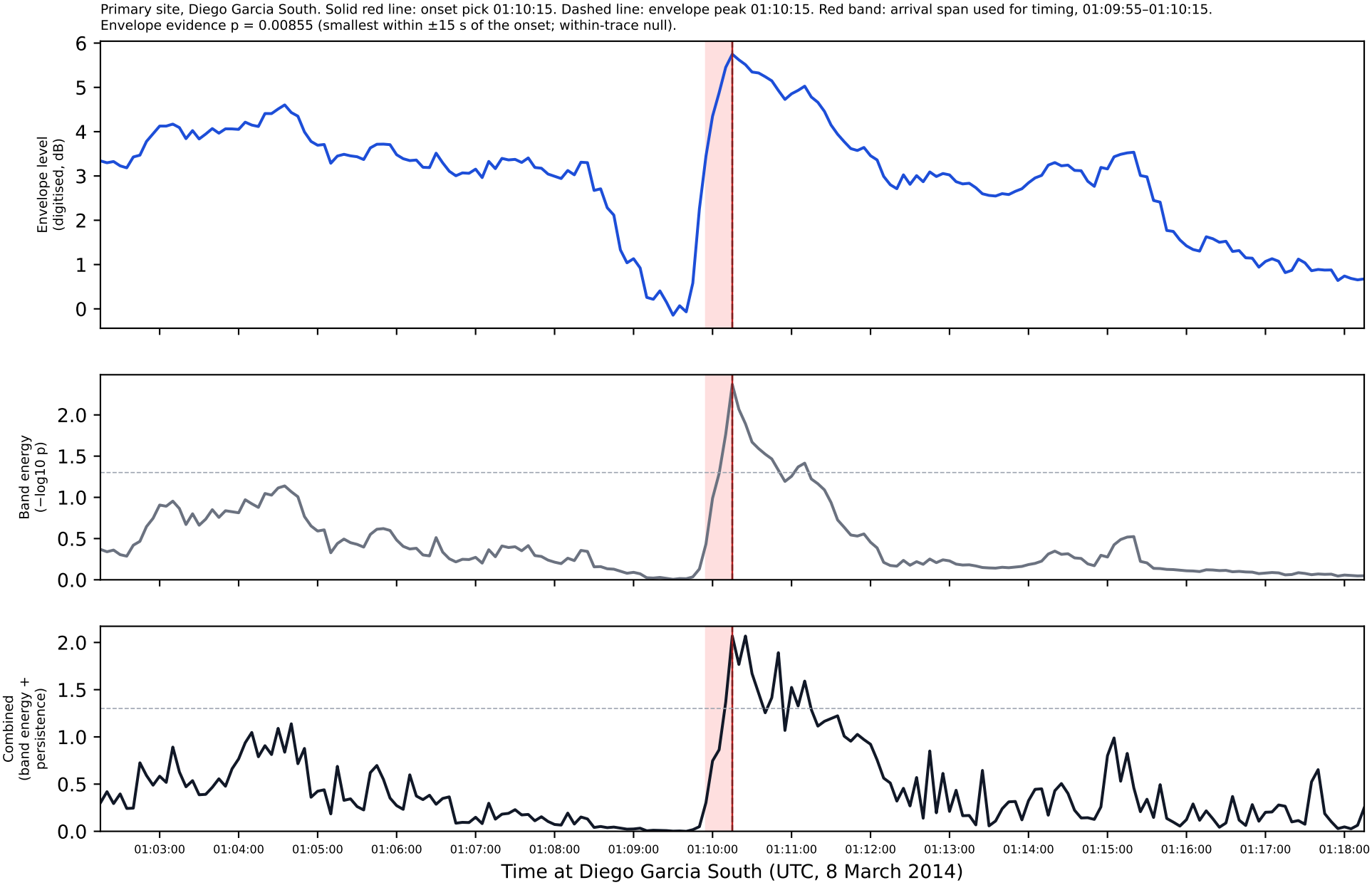
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 2: Cape Leeuwin 00:47:04 (Kadri and ours; bearing 338.7° corrected, 343.2° published). Implied impact 00:19:38–00:27:03 UTC, arc 15; 16° S



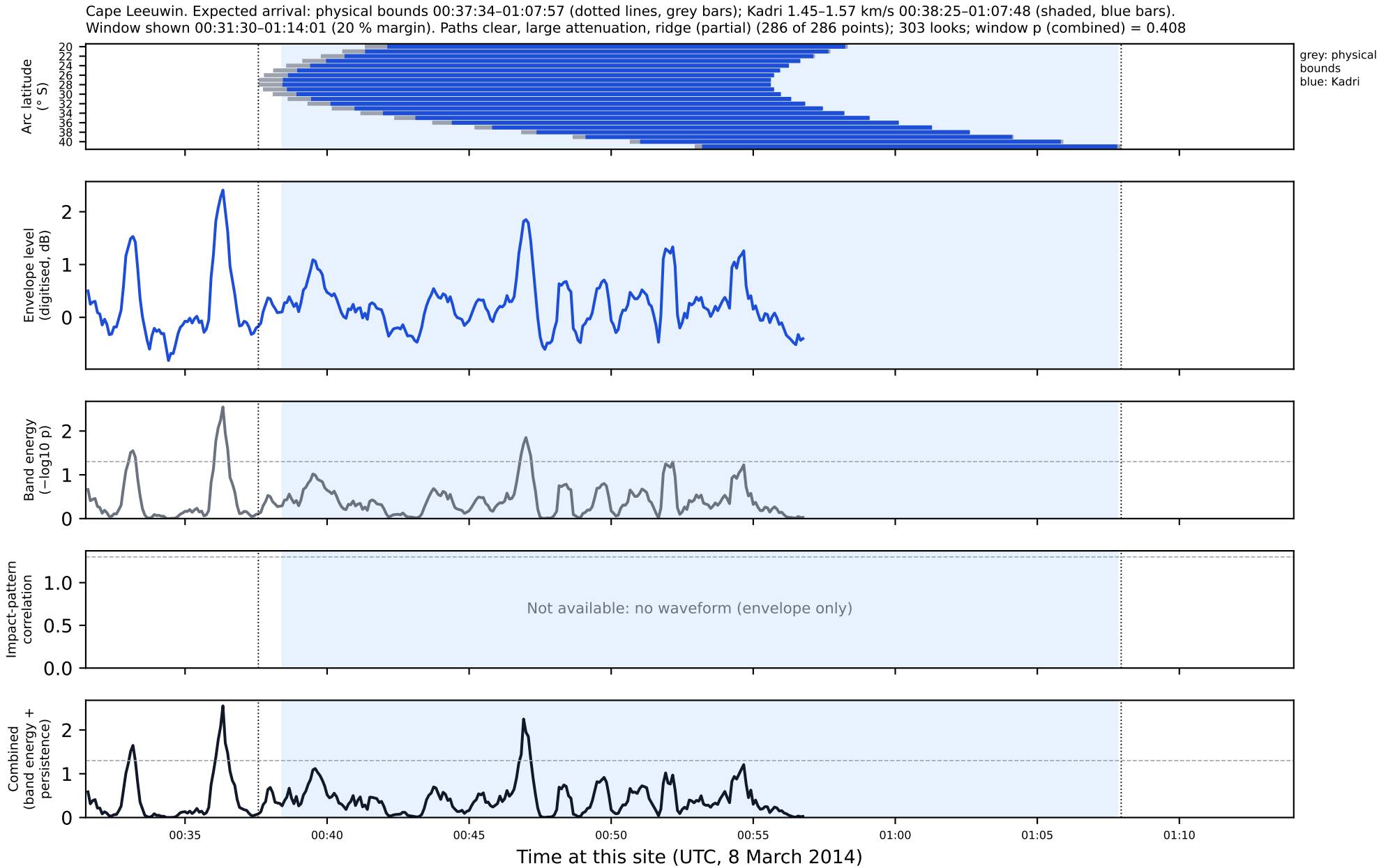
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Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:21:52–00:34:29 UTC, arc latitude from timing only



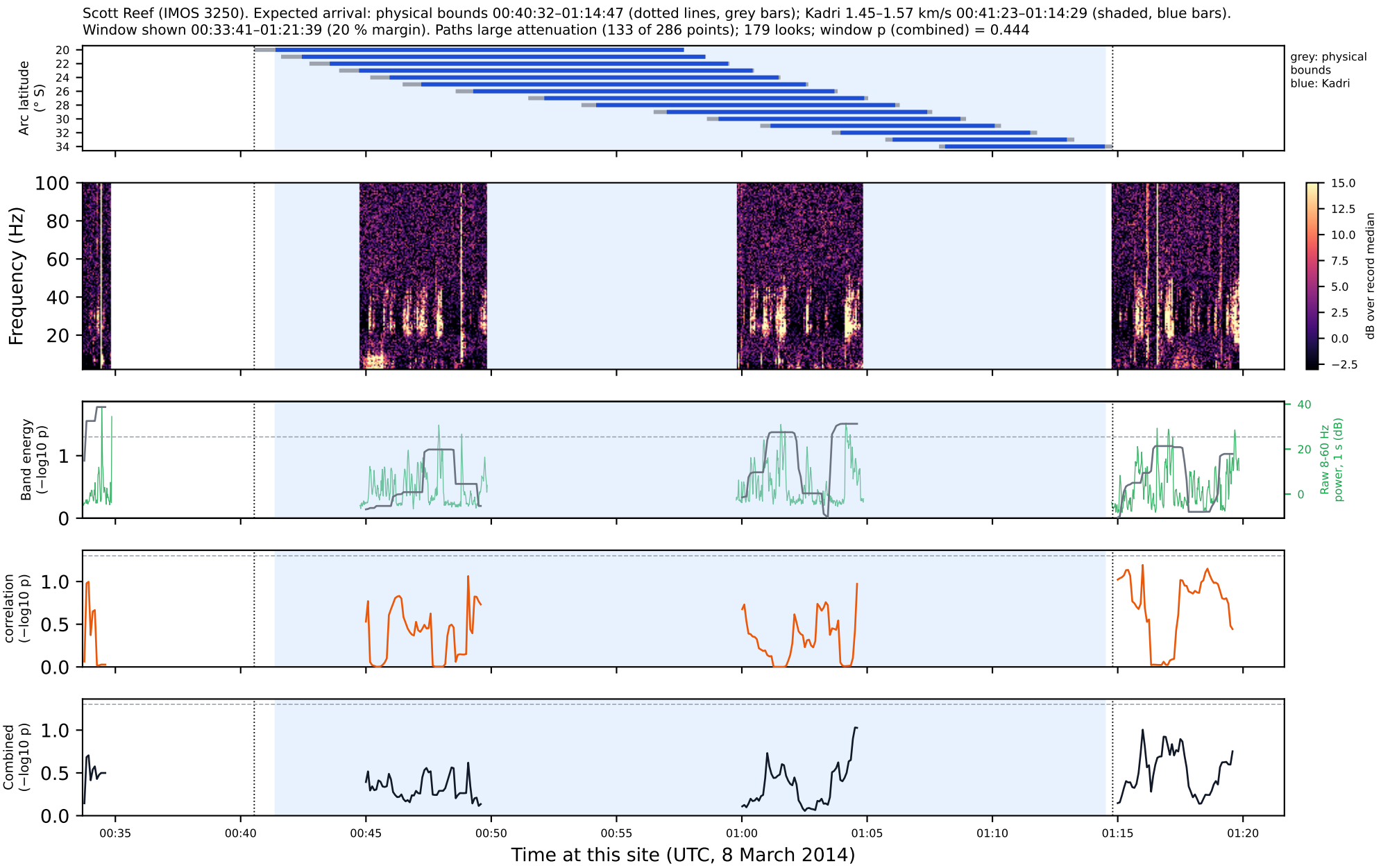
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

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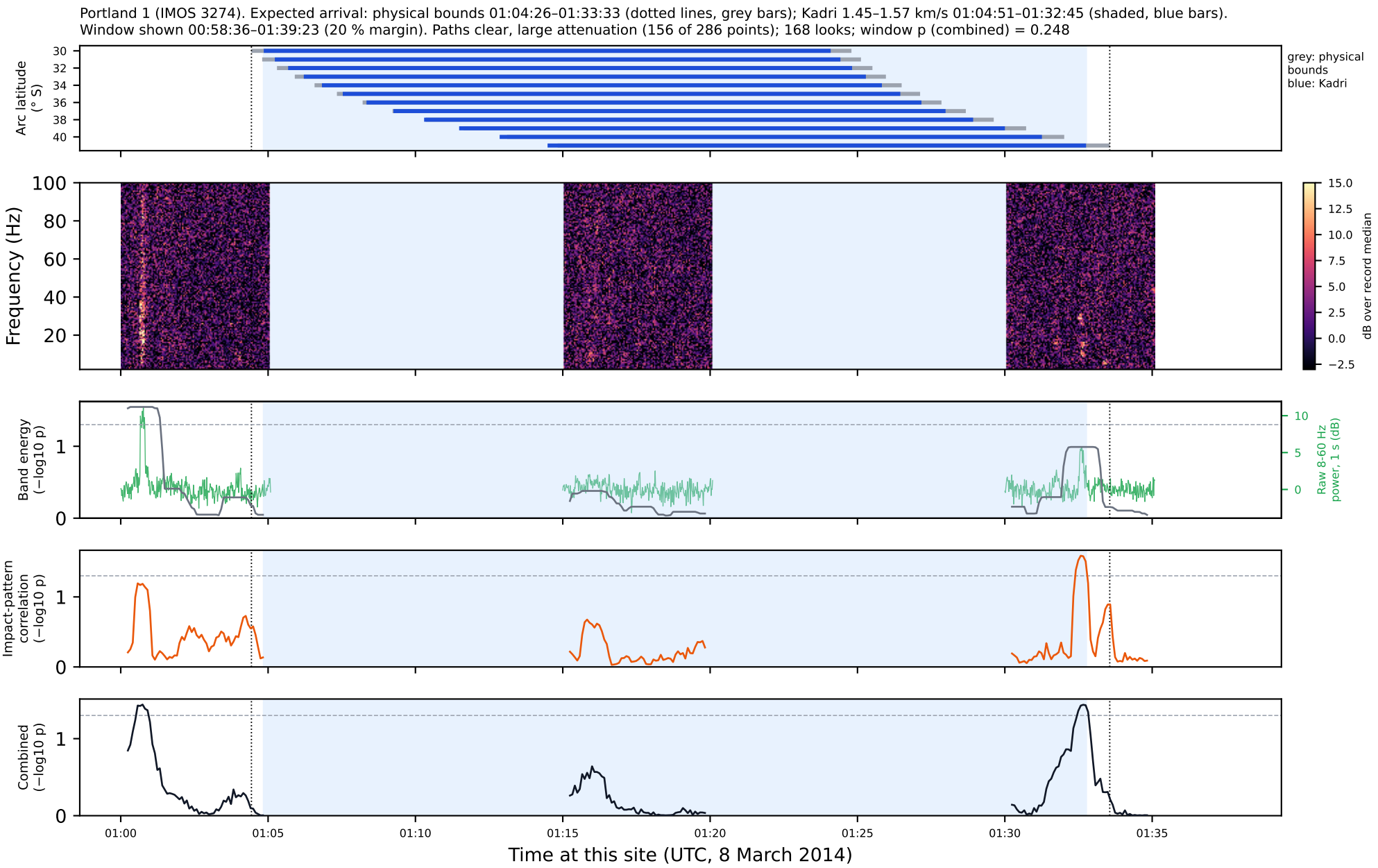
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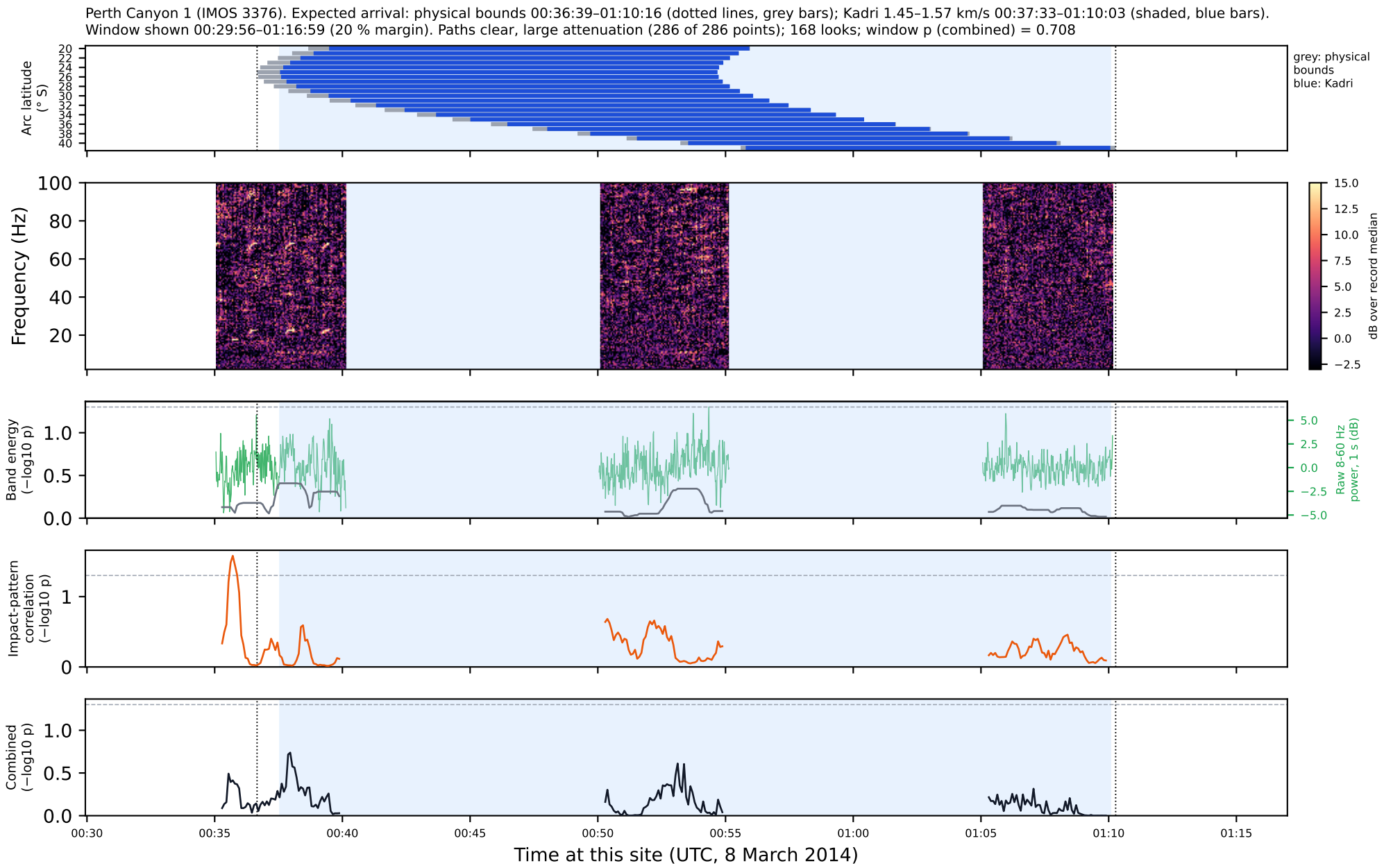
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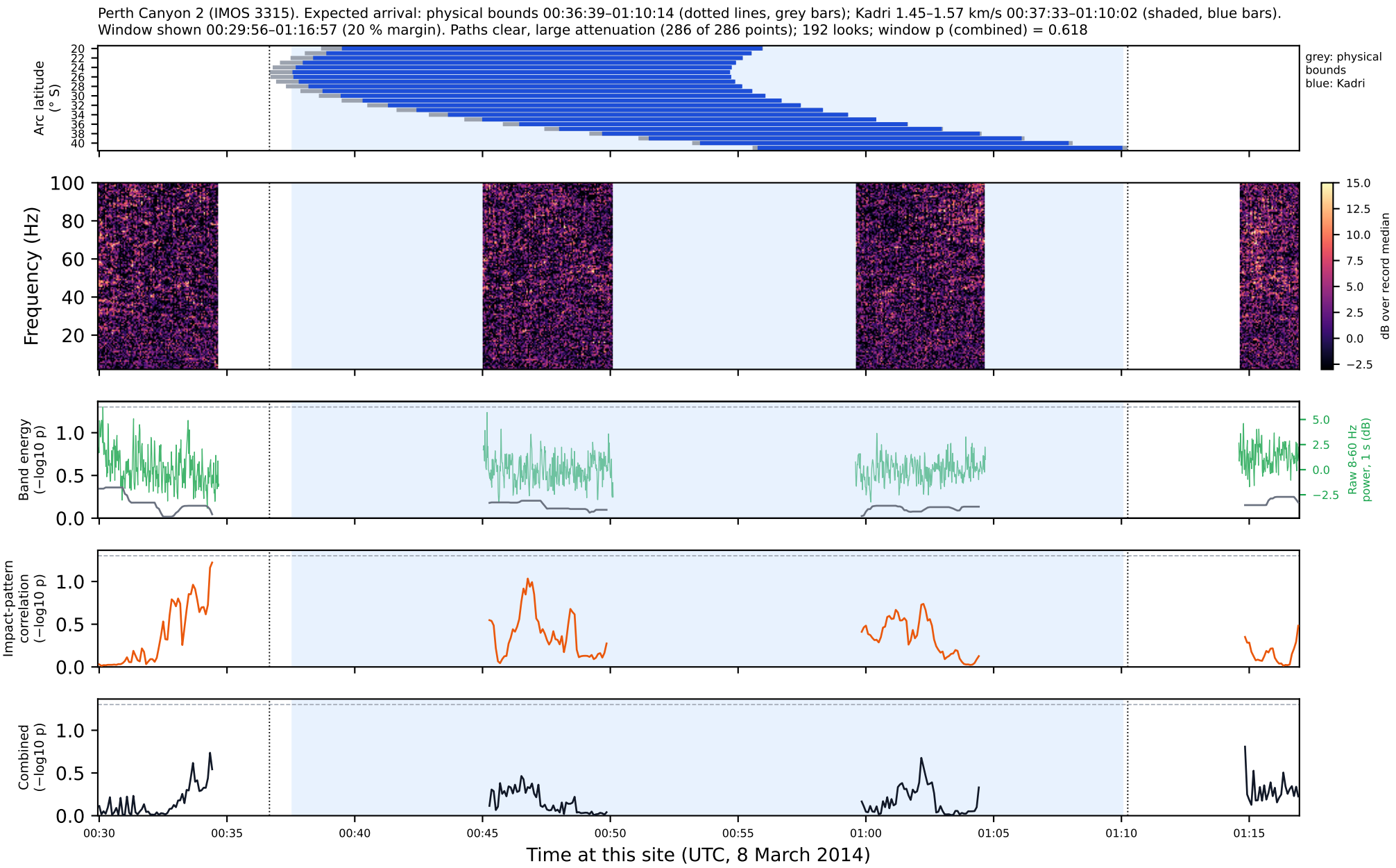
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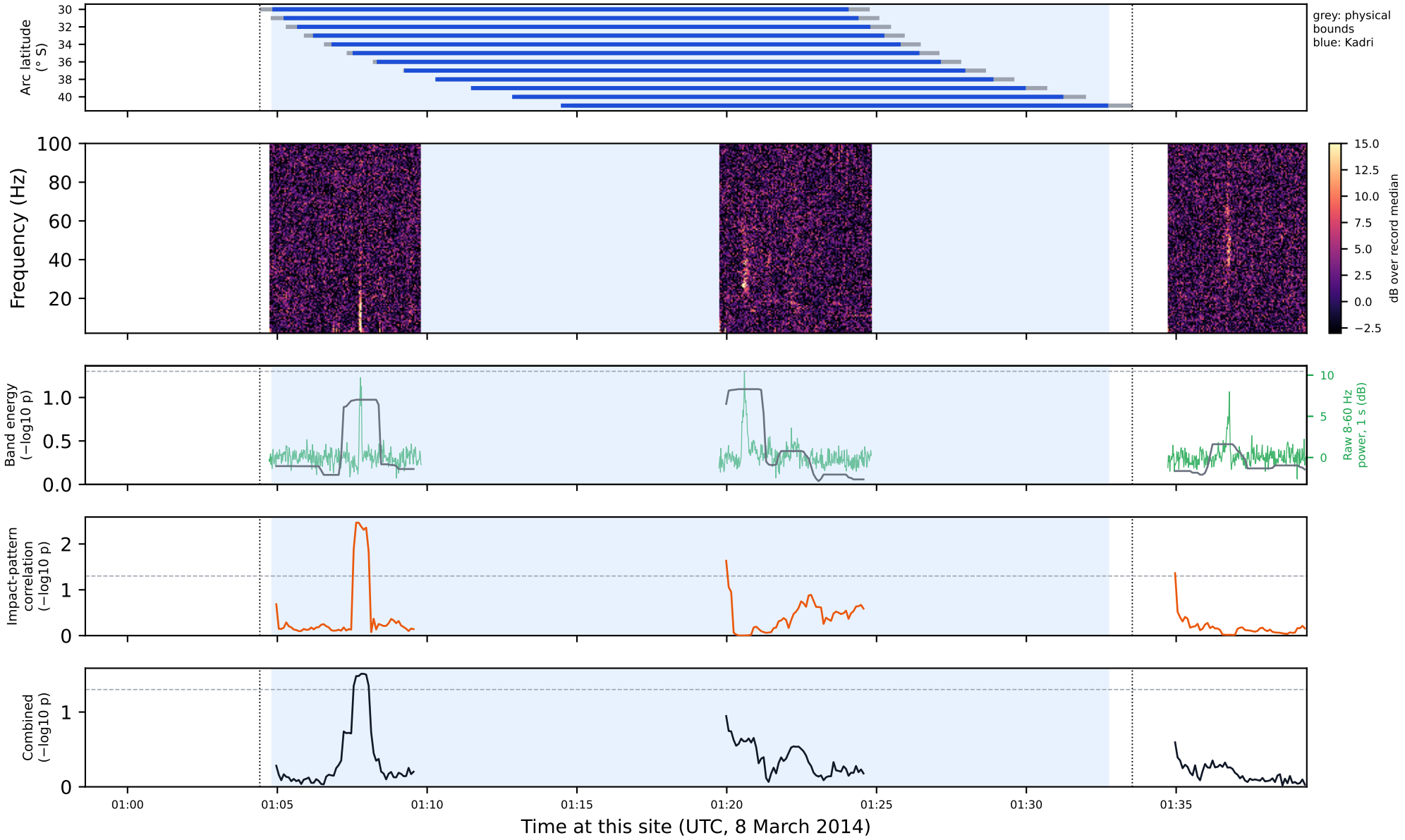
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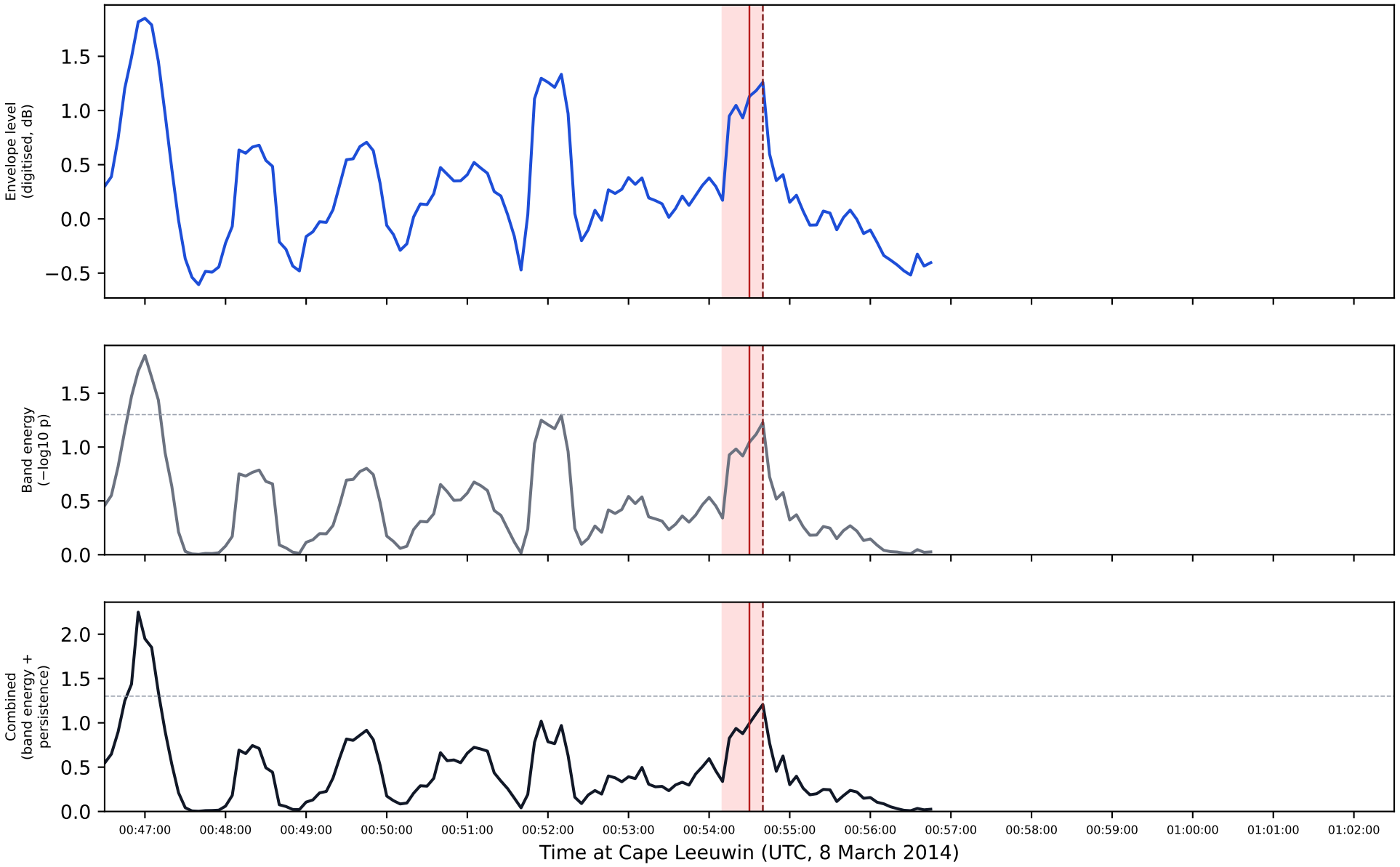
Portland 2 (IMOS 3275). Expected arrival: physical bounds 01:04:24–01:33:32 (dotted lines, grey bars); Kadri 1.45–1.57 km/s 01:04:50–01:32:44 (shaded, blue bars).
Window shown 00:58:35–01:39:21 (20 % margin). Paths clear, large attenuation (156 of 286 points); 165 looks; window p (combined) = 0.176



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

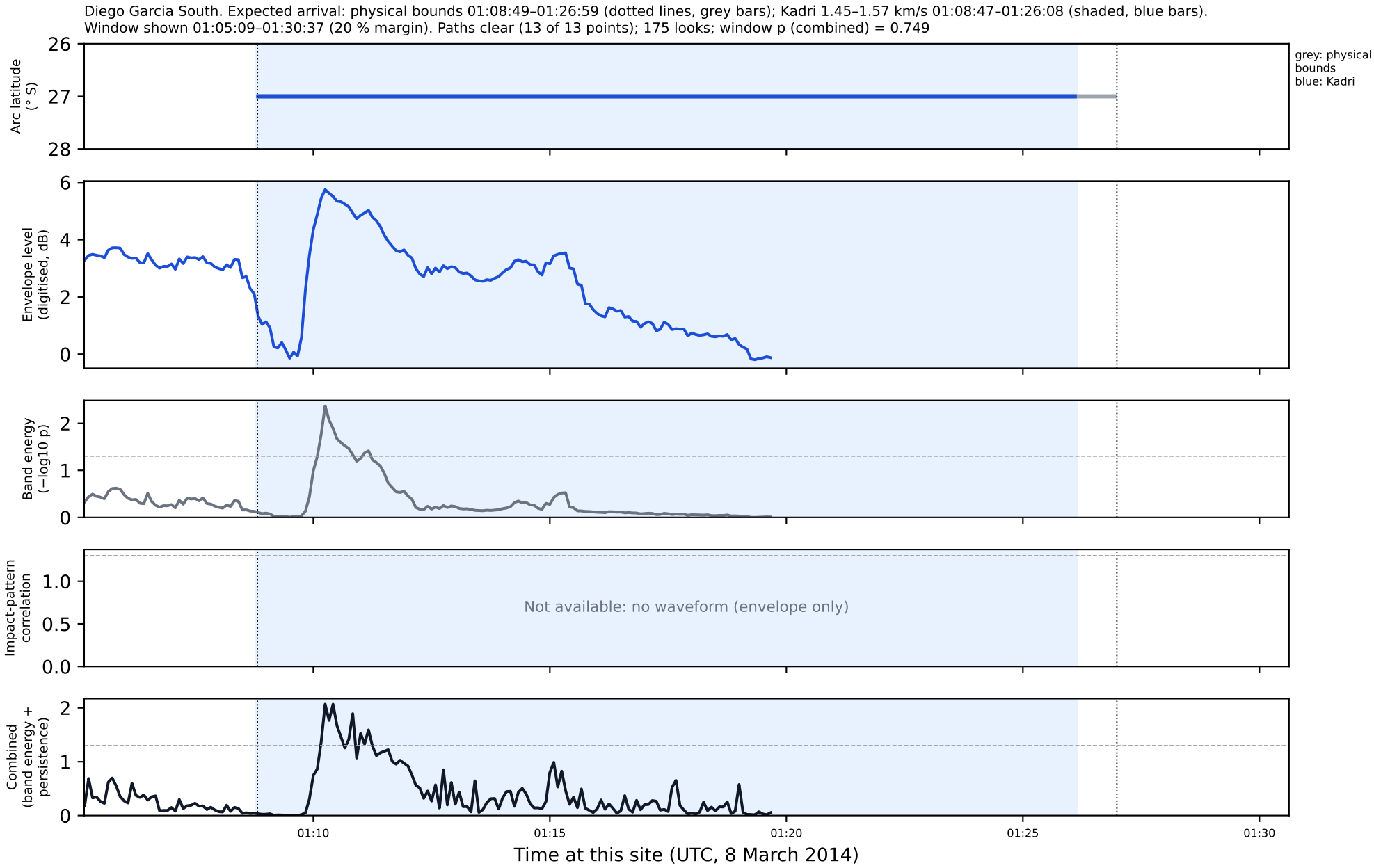
Candidate 4: Cape Leeuwin 00:54:30 (Kadri and ours; bearing 299.4° corrected, 306.2° published). Implied impact 00:32:23–00:40:58 UTC, arc 27° S

Primary site, Cape Leeuwin. Solid red line: onset pick 00:54:30. Dashed line: envelope peak 00:54:40. Red band: arrival span used for timing, 00:54:10–00:54:40.
Envelope evidence p = 0.062 (smallest within ±15 s of the onset; within-trace null).
Coincides in time with Royer 5 (uncertain), table pick -18 s; Royer 5 (uncertain), table pick -20 s (label only, not removed; Royer's event locations rest on his own unverified pick association).



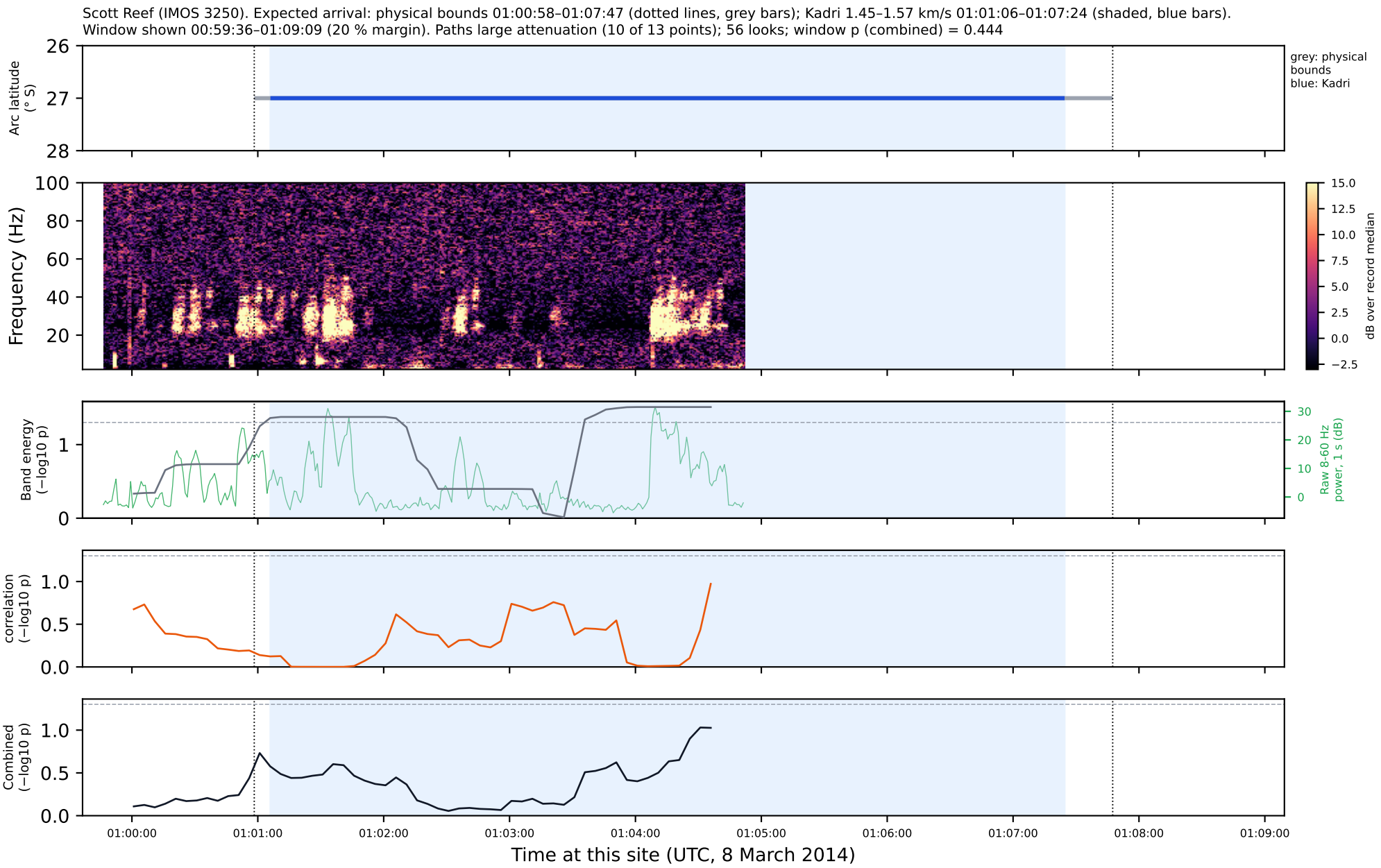
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

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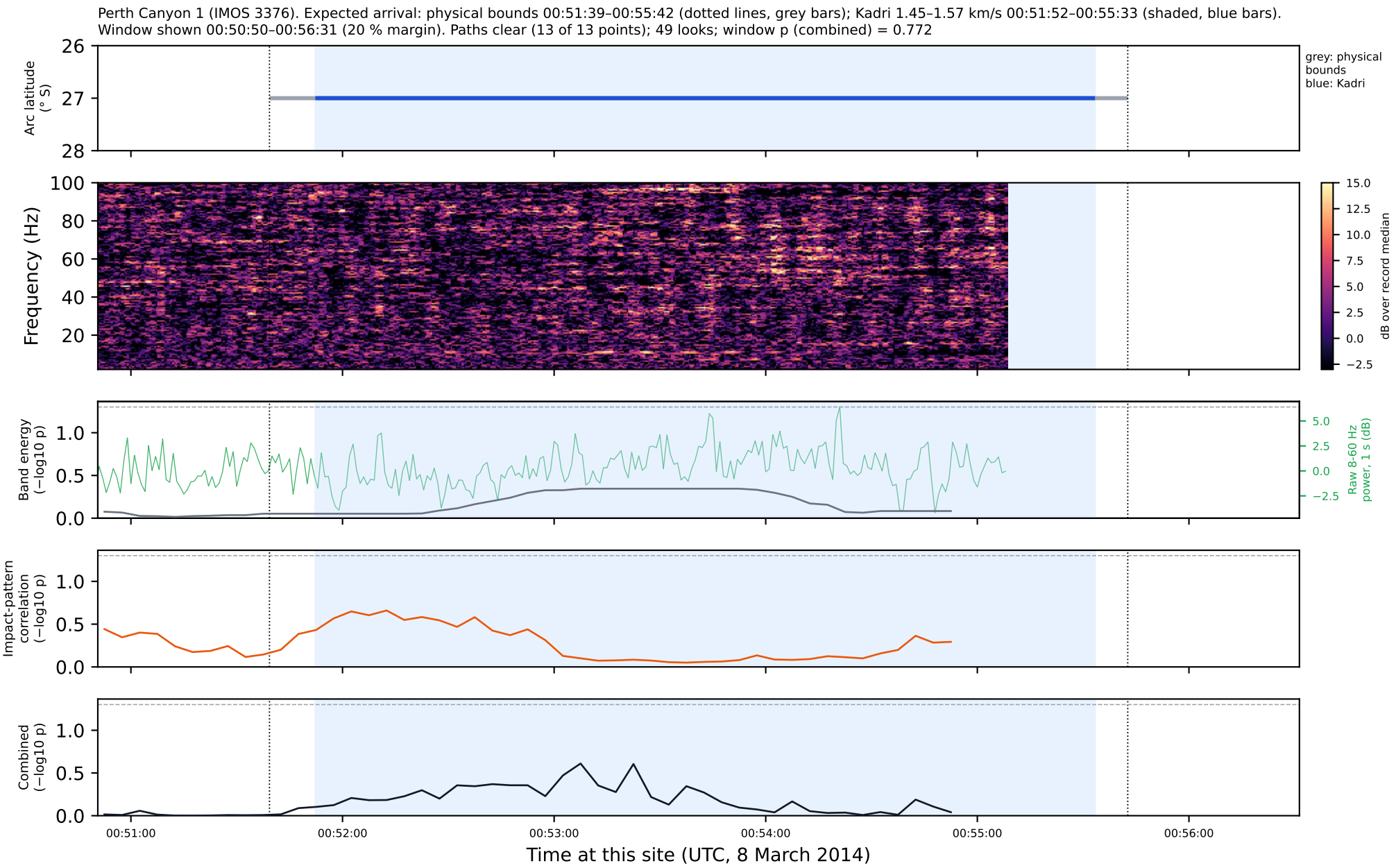
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Candidate 4: Cape Leeuwin 00:54:30 (Kadri and ours; bearing 299.4° corrected, 306.2° published). Implied impact 00:32:23–00:40:58 UTC, arc 27° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

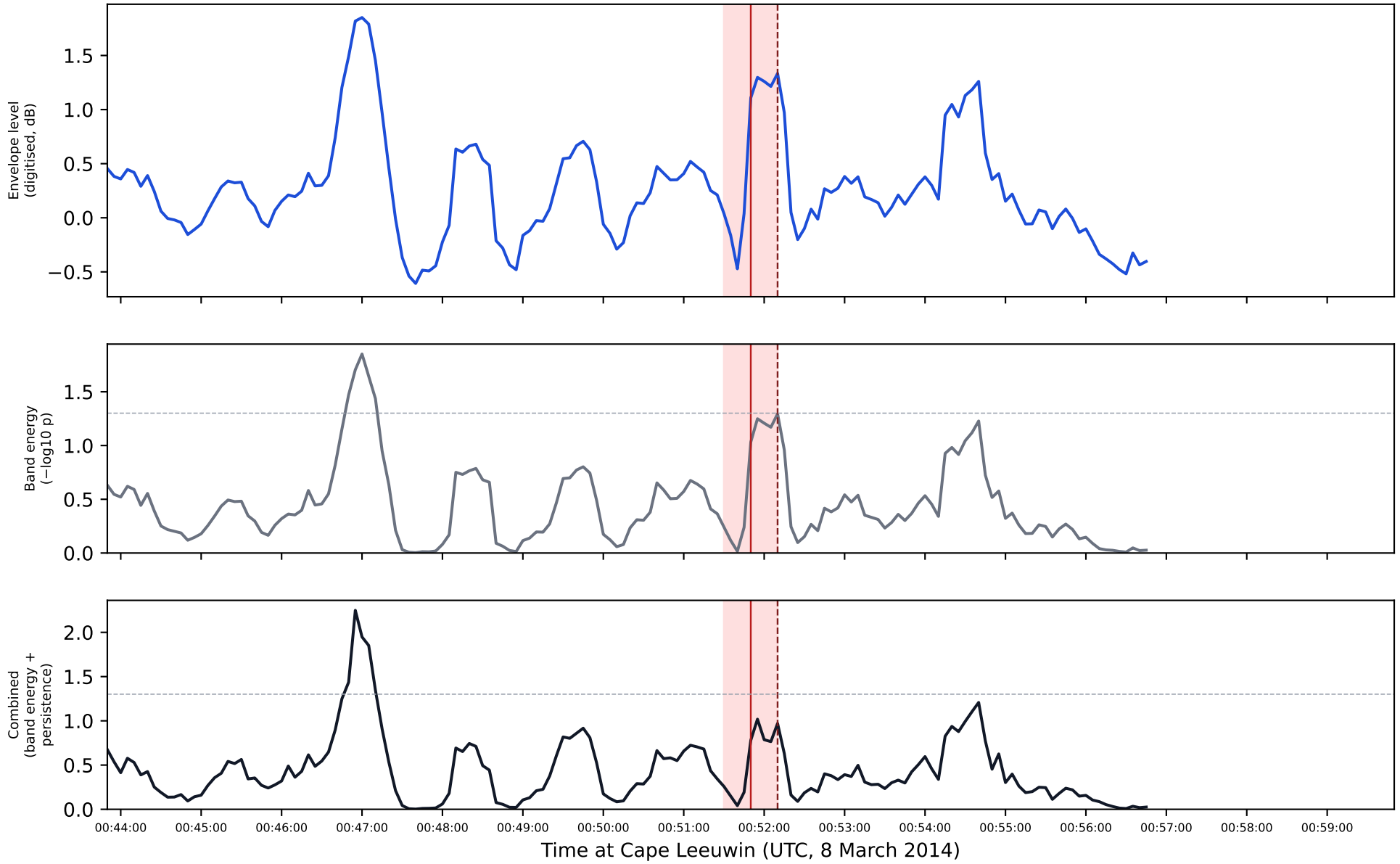
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Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

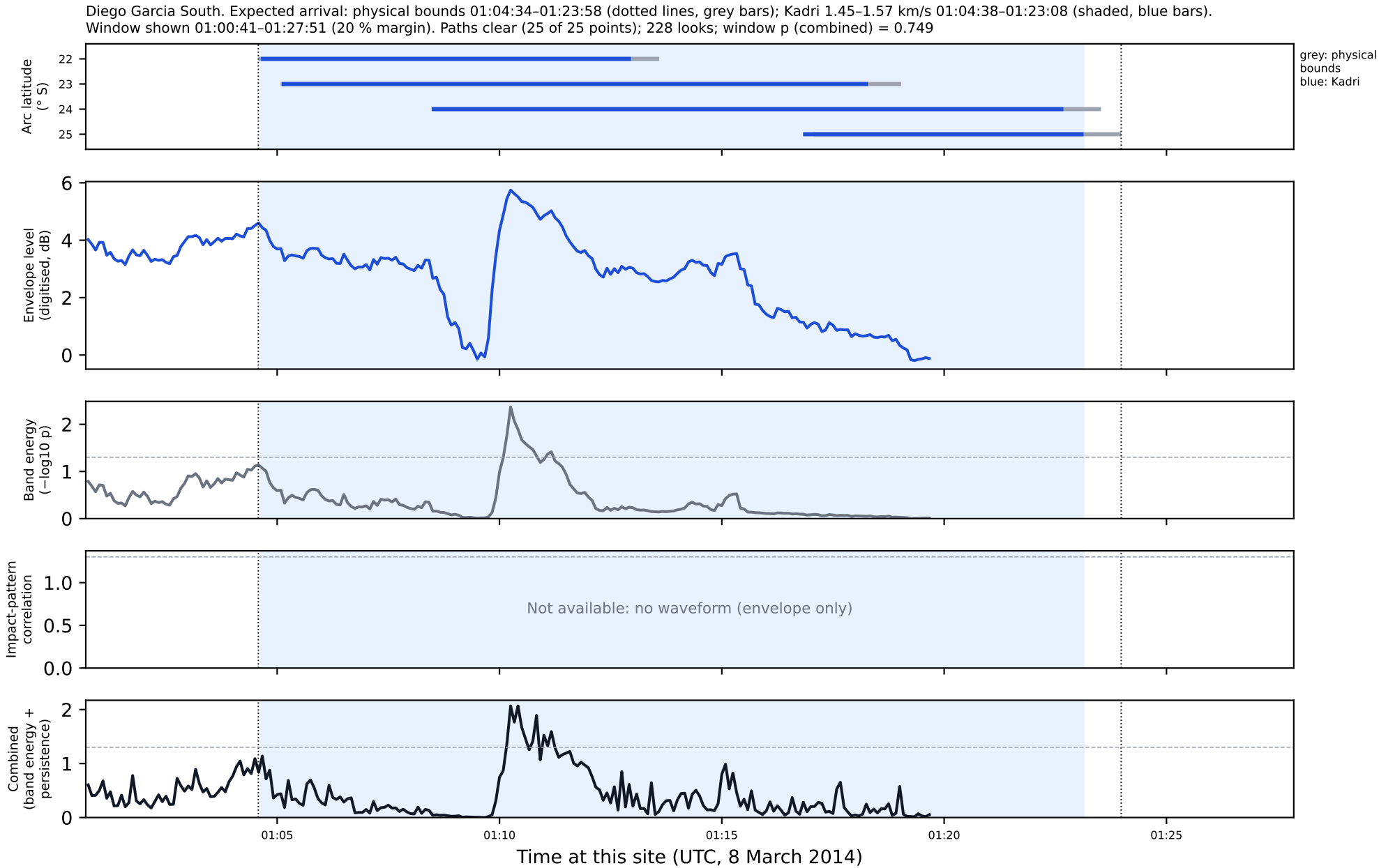
Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:27:47–00:37:50 UTC, arc 23; 24° S

Primary site, Cape Leeuwin. Solid red line: onset pick 00:51:50. Dashed line: envelope peak 00:52:10. Red band: arrival span used for timing, 00:51:30–00:52:10.
Envelope evidence p = 0.0958 (smallest within ±15 s of the onset; within-trace null).
Coincides in time with Royer A, figure pick +7 s; Royer A (icequake), table pick -7 s (label only, not removed; Royer's event locations rest on his own unverified pick association).



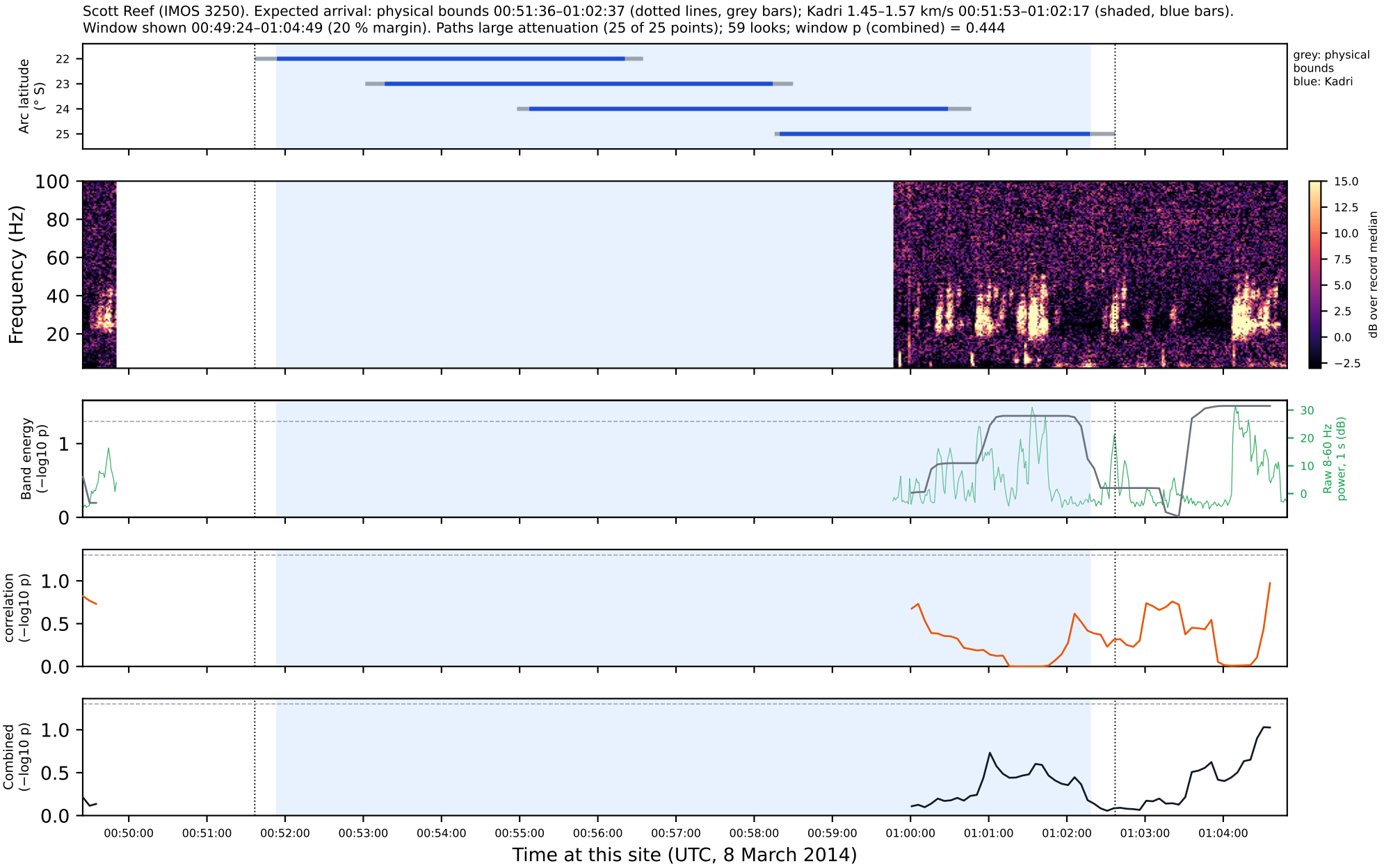
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

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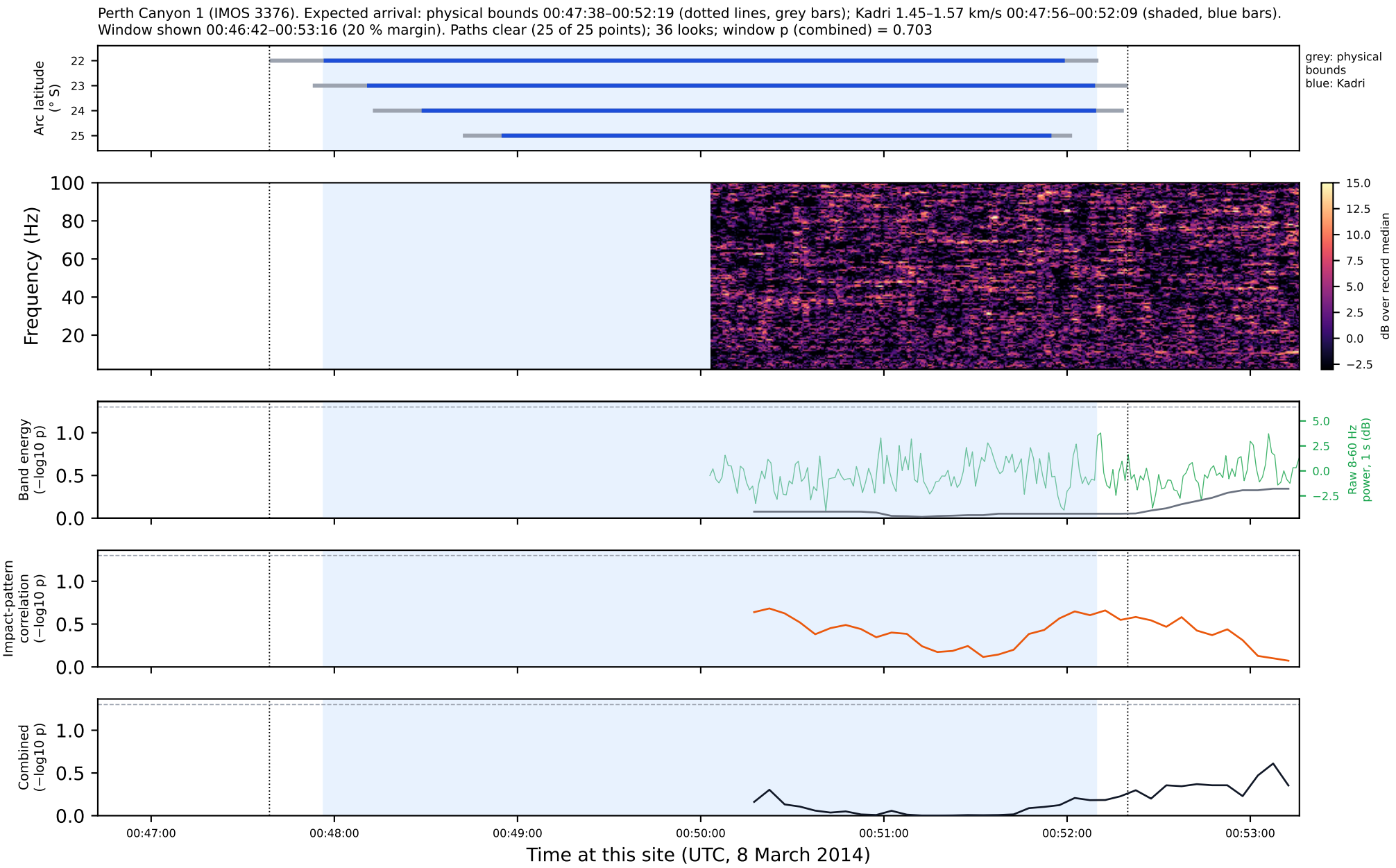
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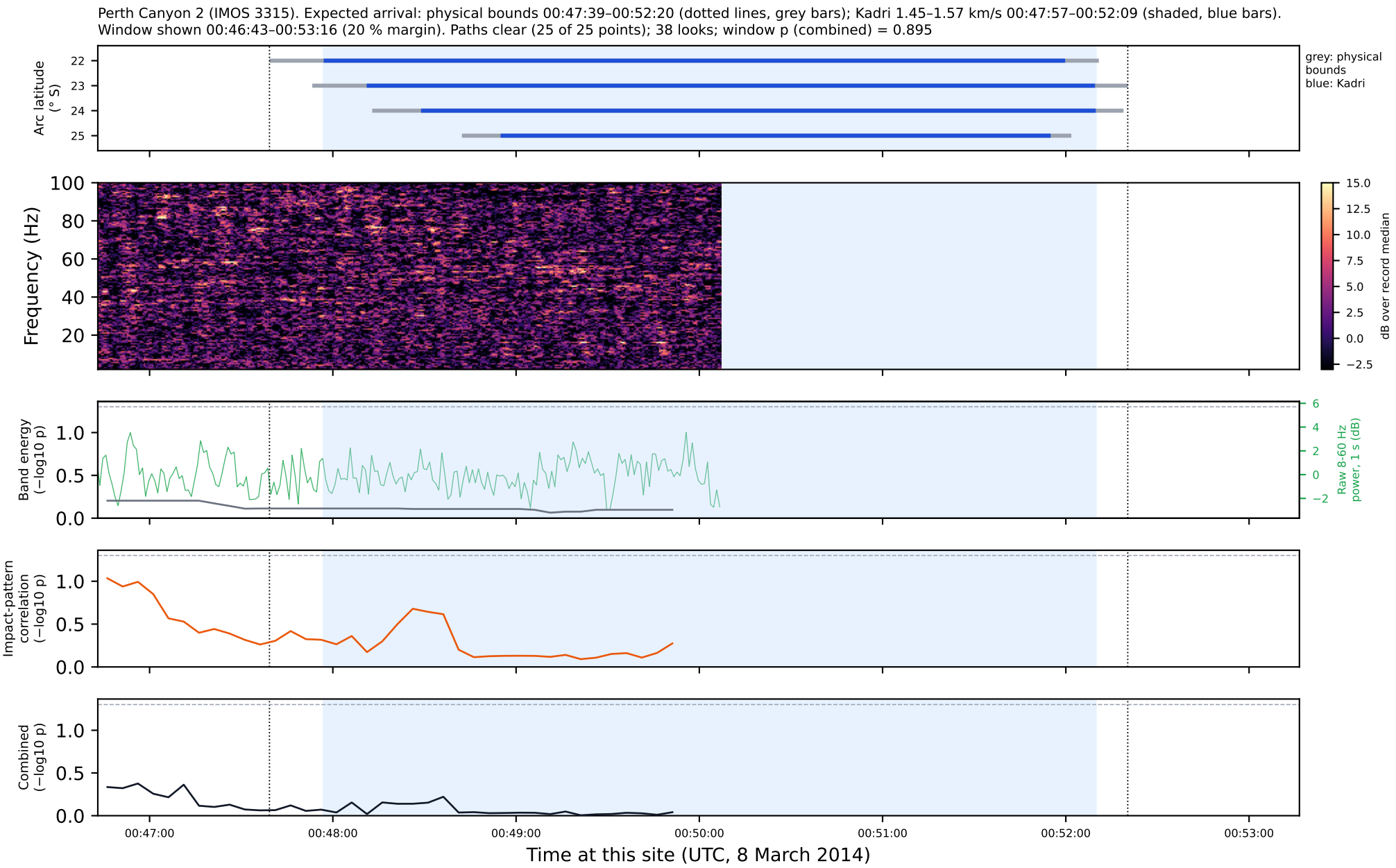
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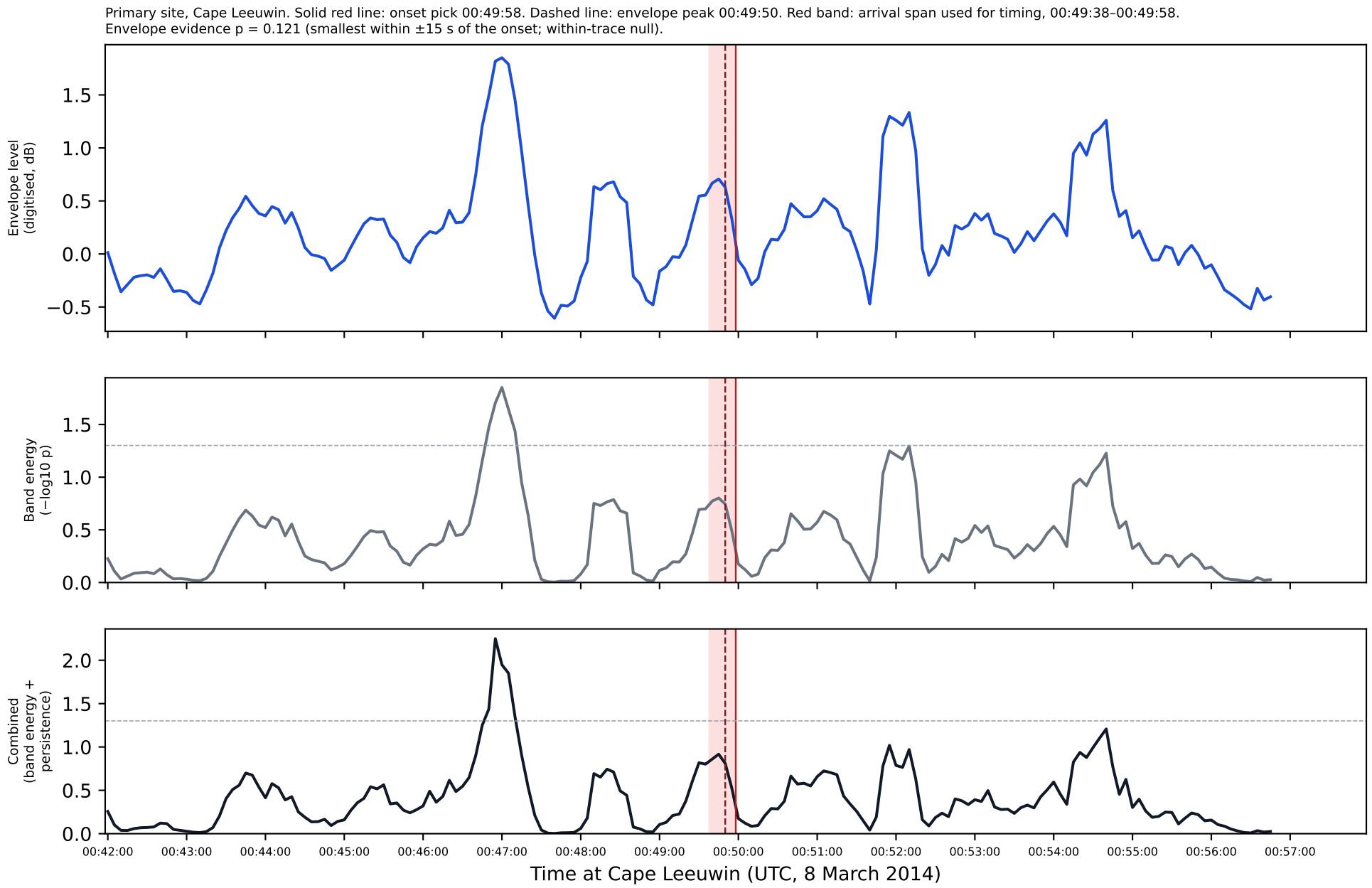
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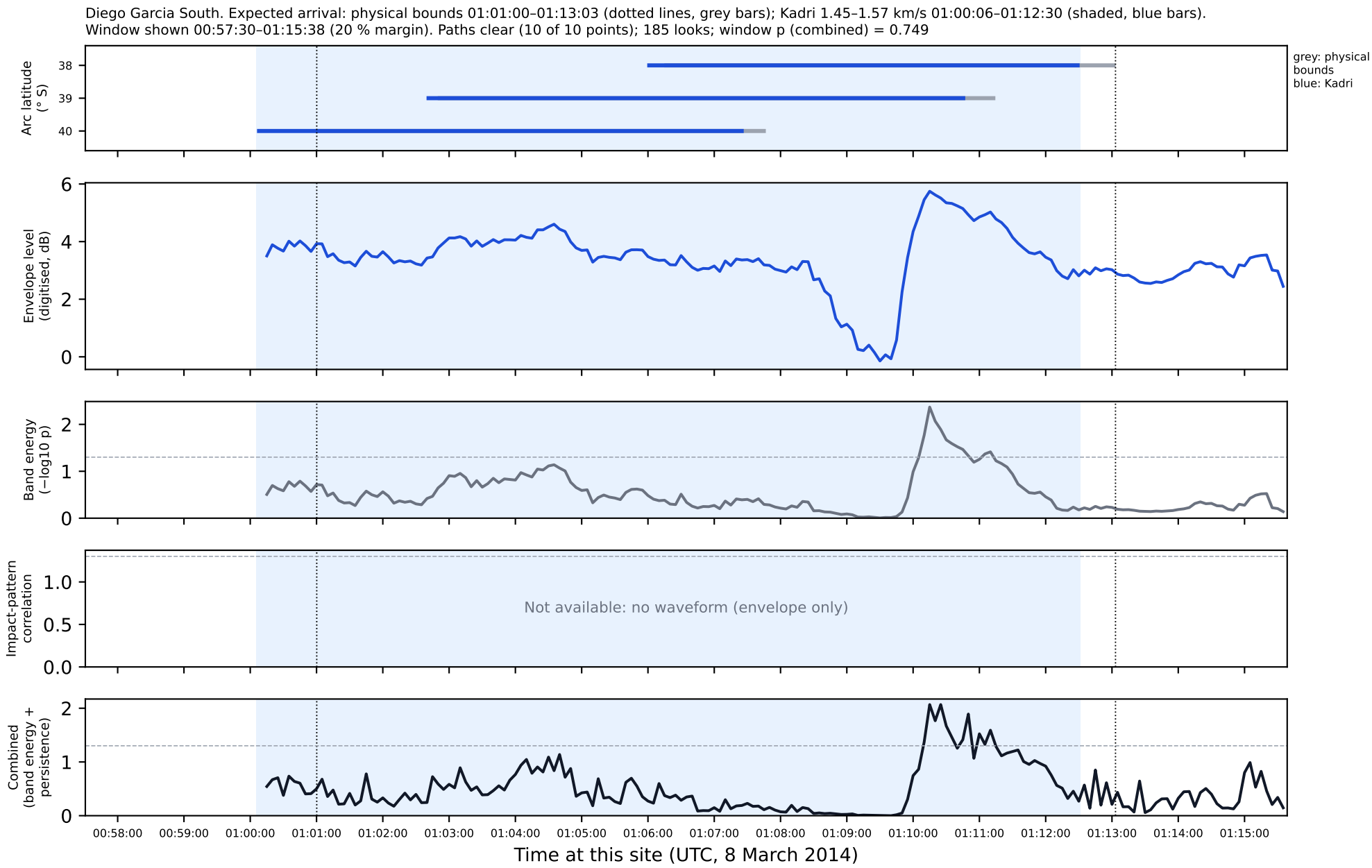
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Candidate 6: Cape Leeuwin 00:49:58 (Kadri; bearing 246.8° corrected, 260.4° published). Implied impact 00:19:38–00:26:37 UTC, arc 38; 39; 40° S



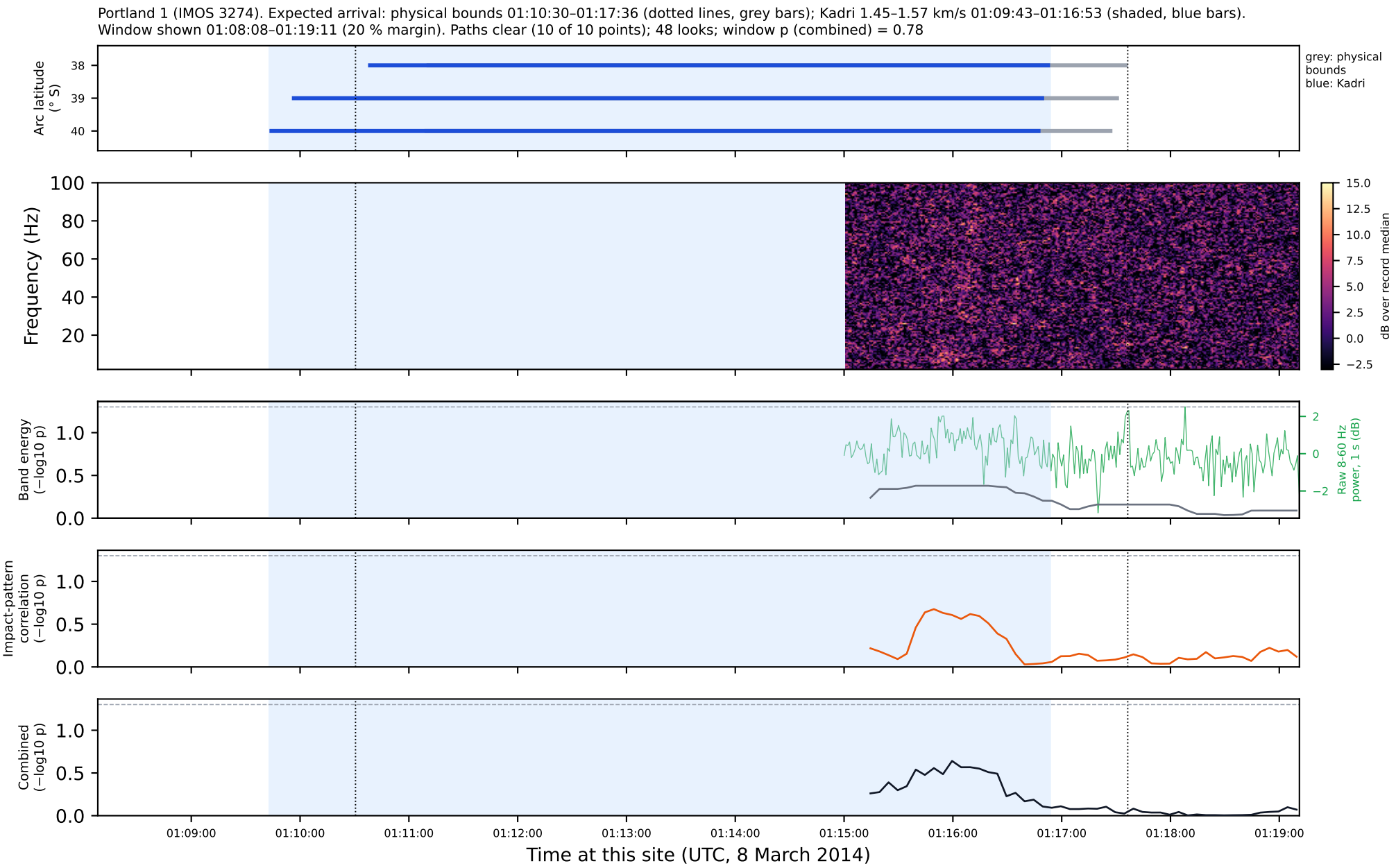
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

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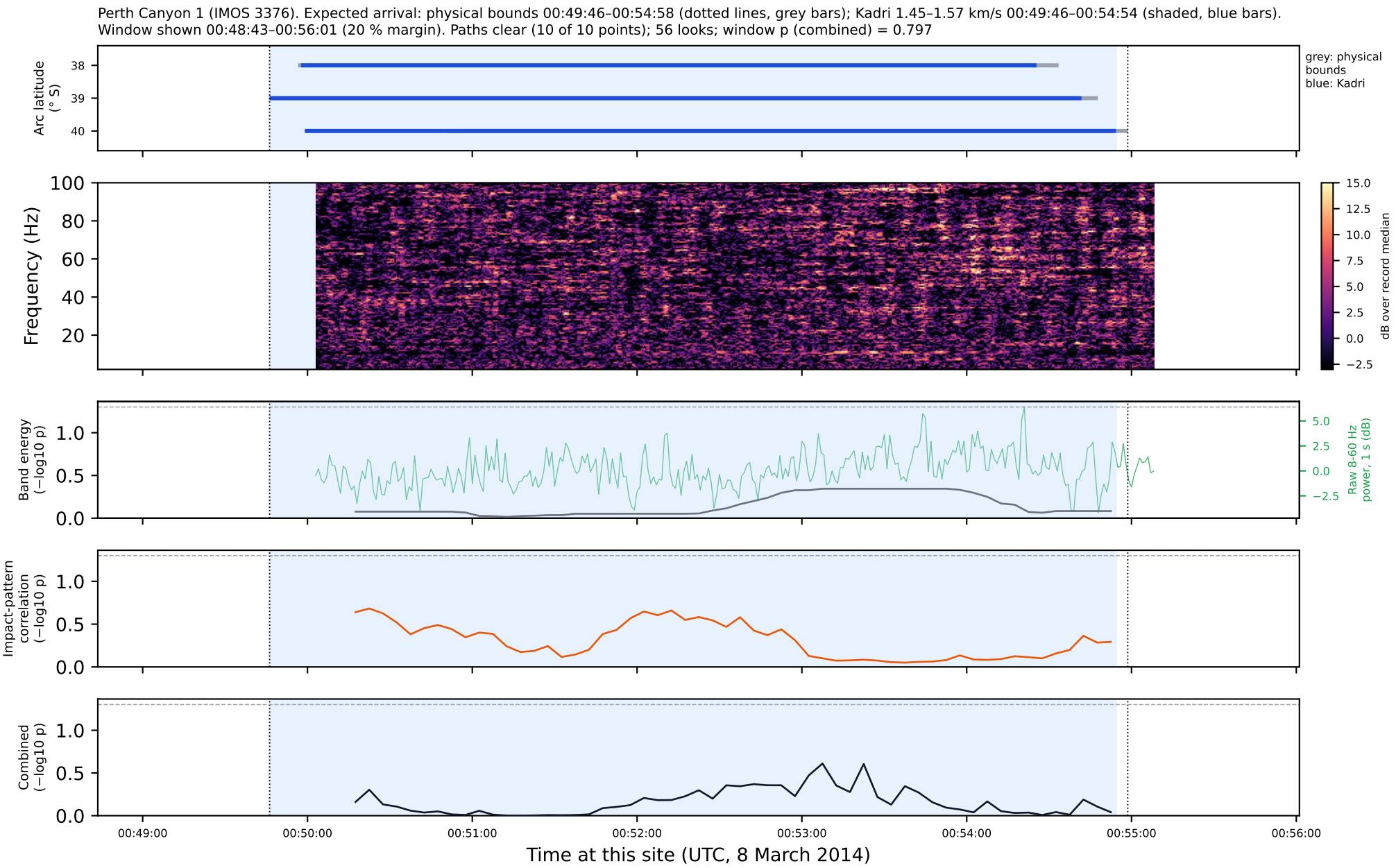
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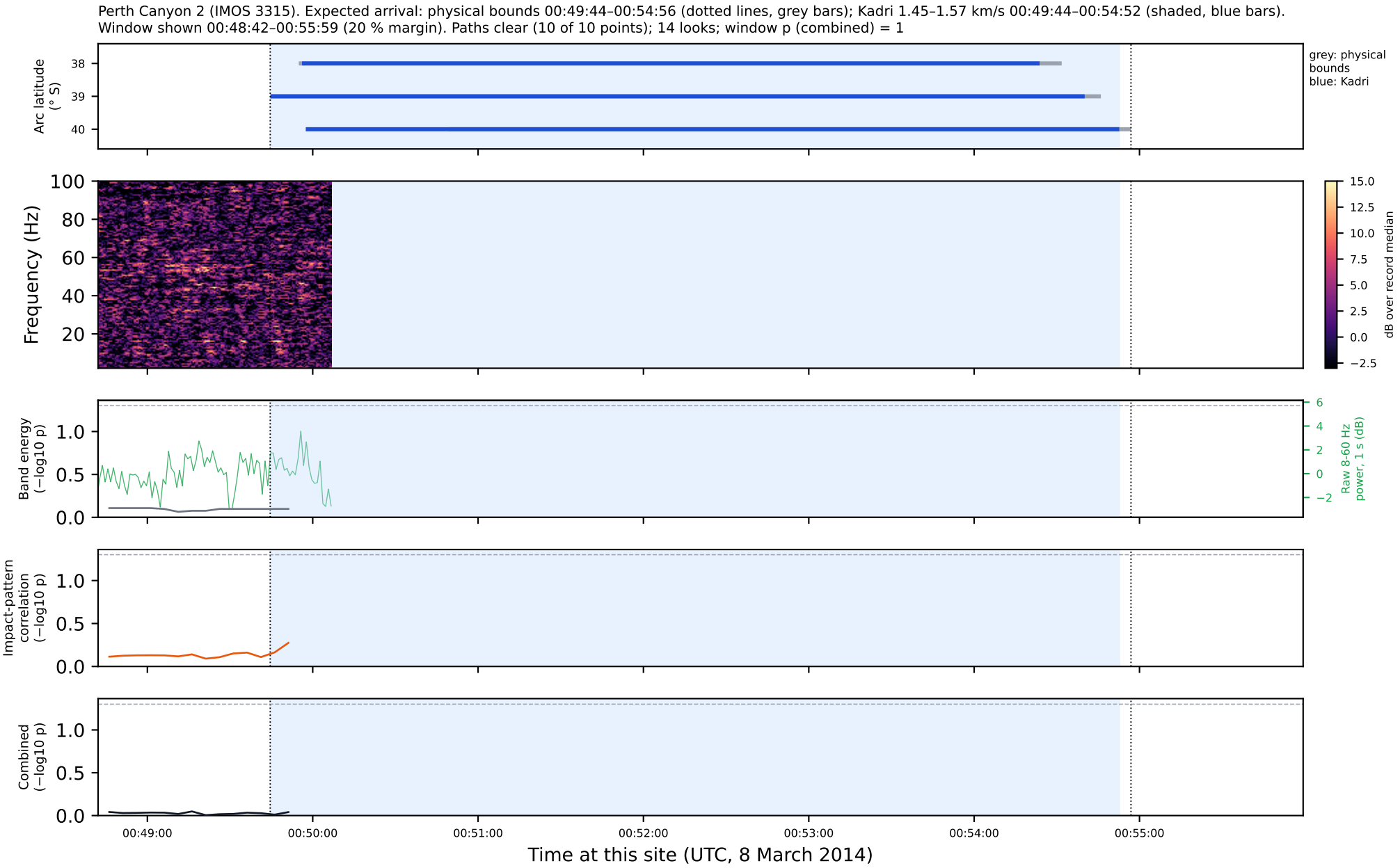
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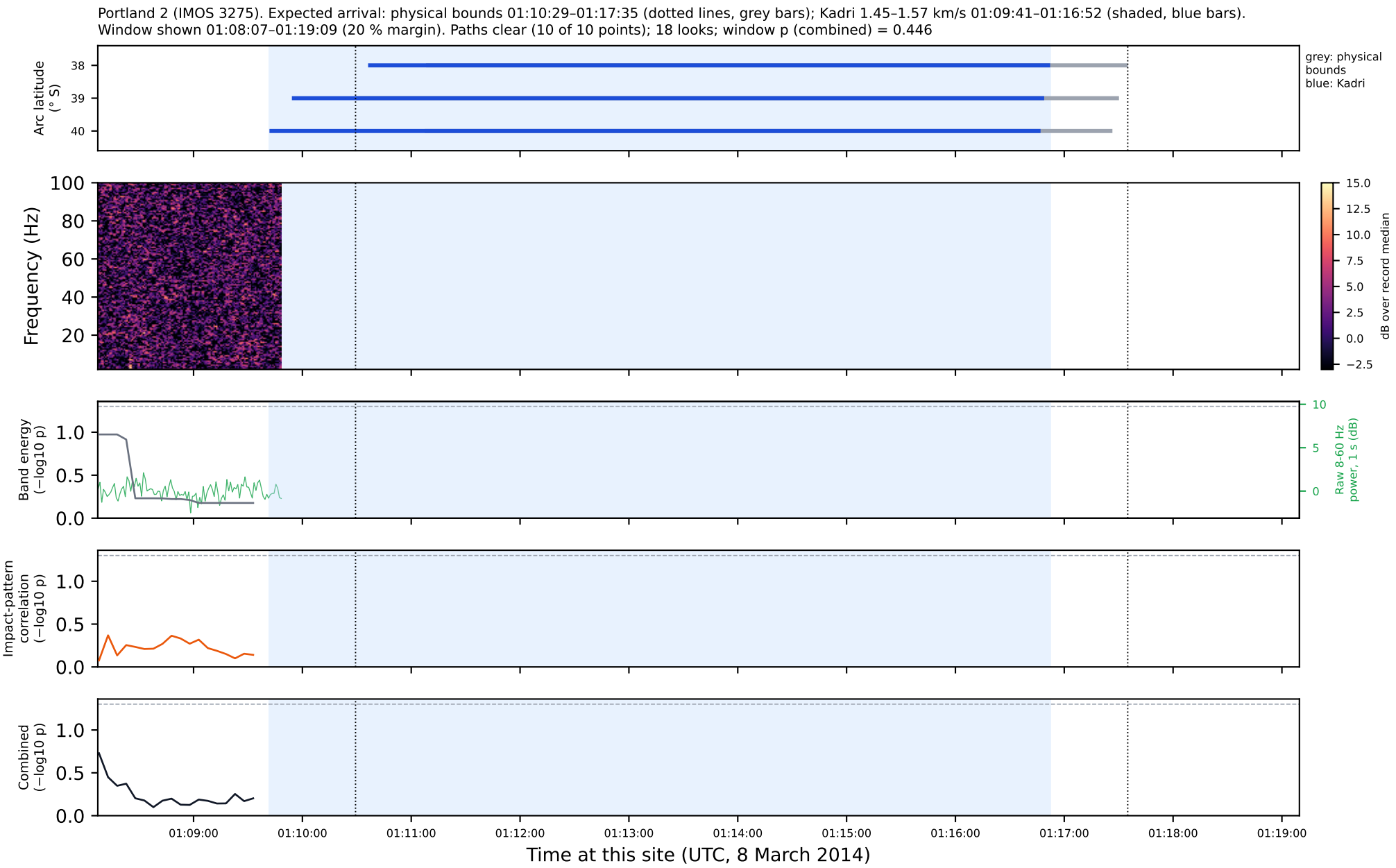
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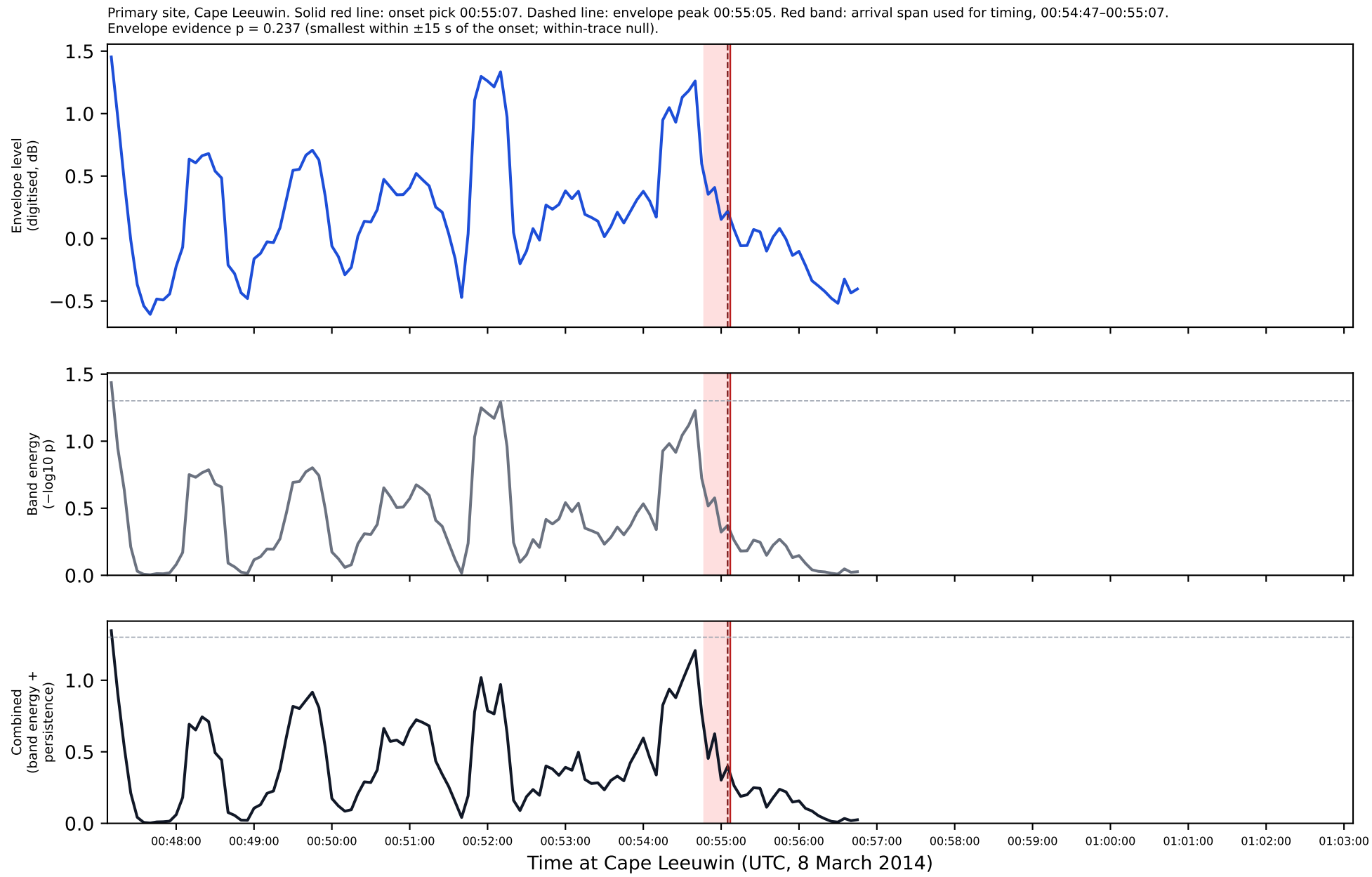
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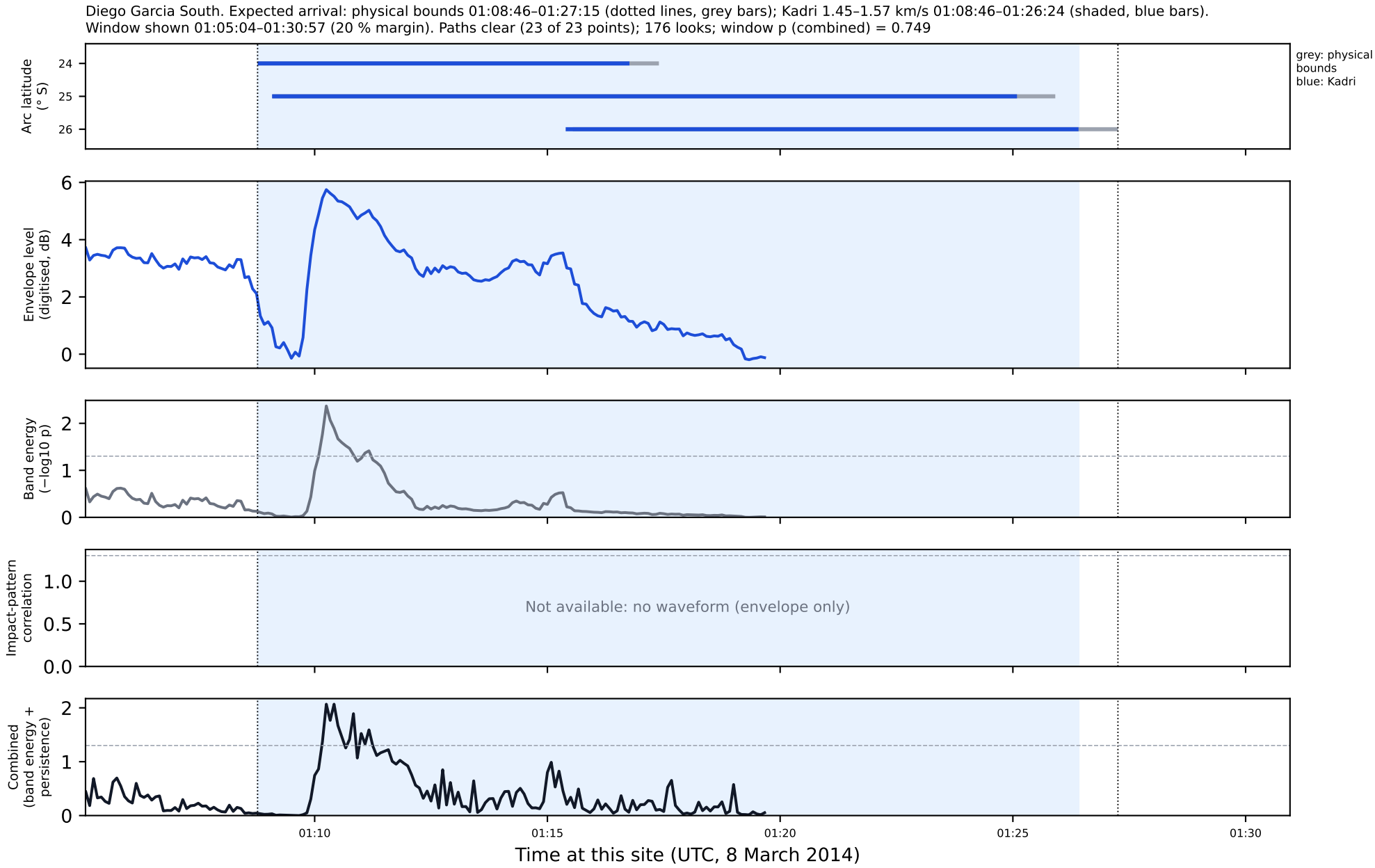
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Candidate 7: Cape Leeuwin 00:55:07 (Kadri; bearing 308.5° corrected, 314.1° published). Implied impact 00:32:09–00:41:11 UTC, arc 25; 26° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

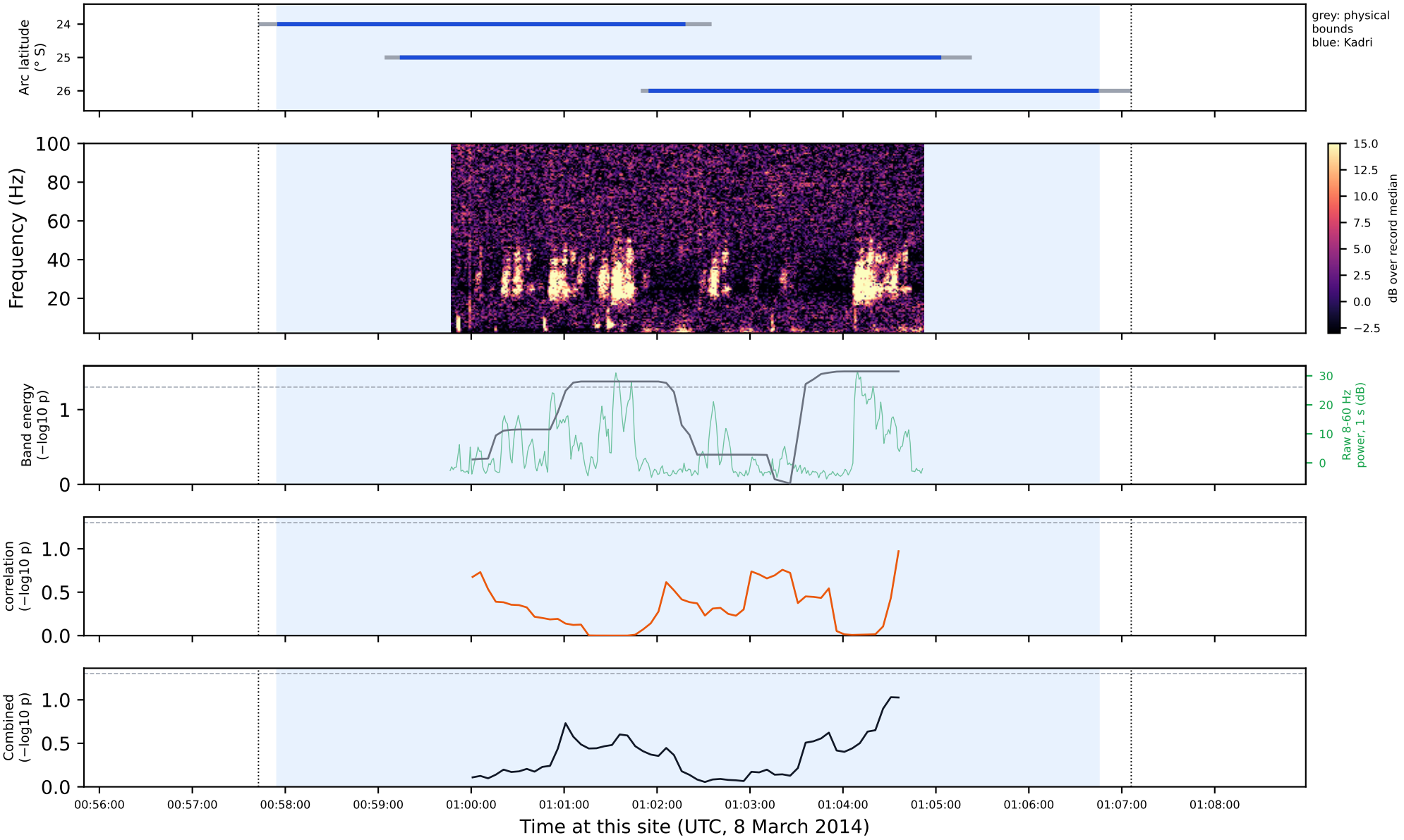
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Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

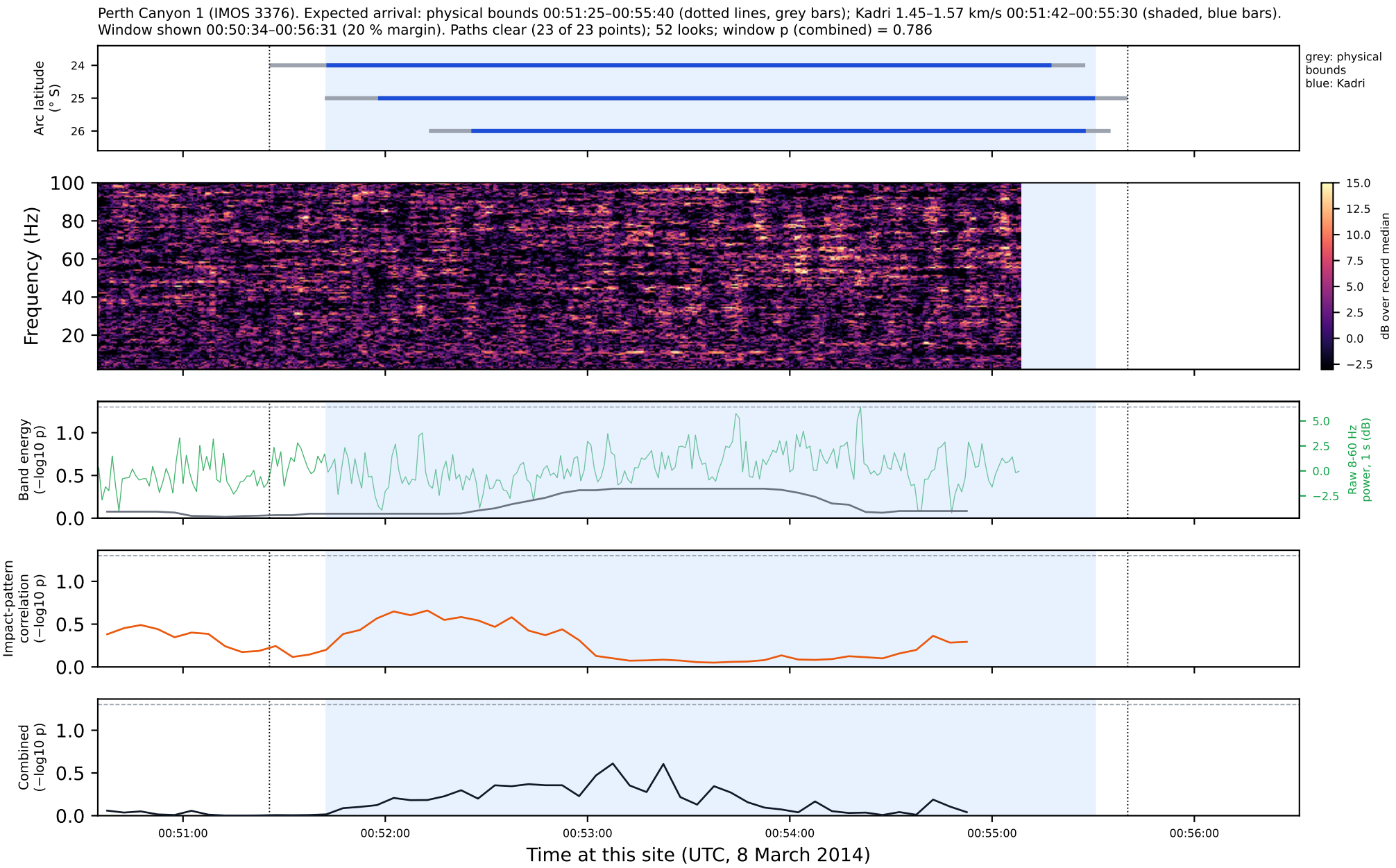
Candidate 7: Cape Leeuwin 00:55:07 (Kadri; bearing 308.5° corrected, 314.1° published). Implied impact 00:32:09–00:41:11 UTC, arc 25; 26° S

Scott Reef (IMOS 3250). Expected arrival: physical bounds 00:57:42–01:07:06 (dotted lines, grey bars); Kadri 1.45–1.57 km/s 00:57:54–01:06:45 (shaded, blue bars).
Window shown 00:55:50–01:08:58 (20 % margin). Paths large attenuation (22 of 23 points); 56 looks; window p (combined) = 0.444



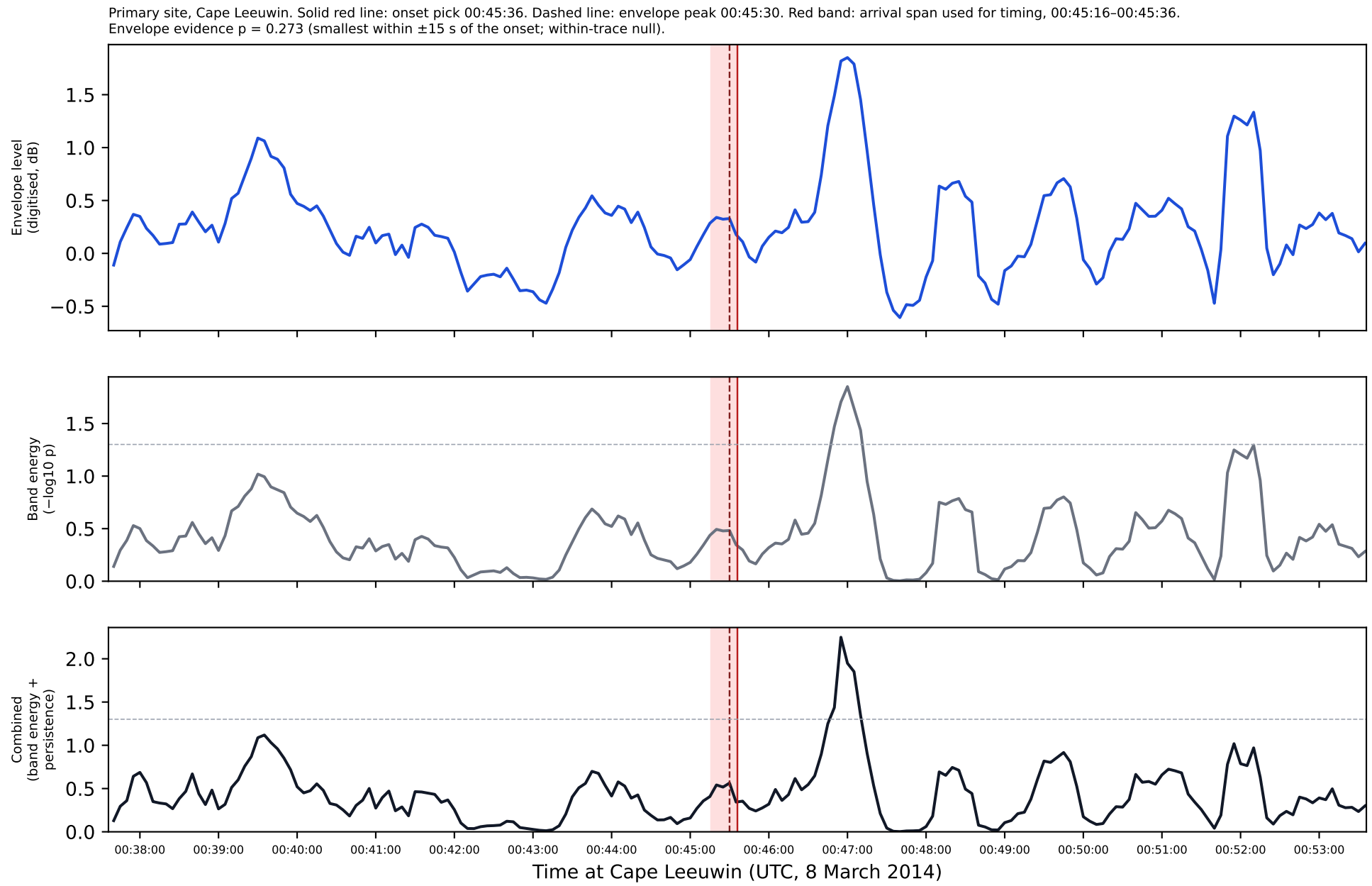
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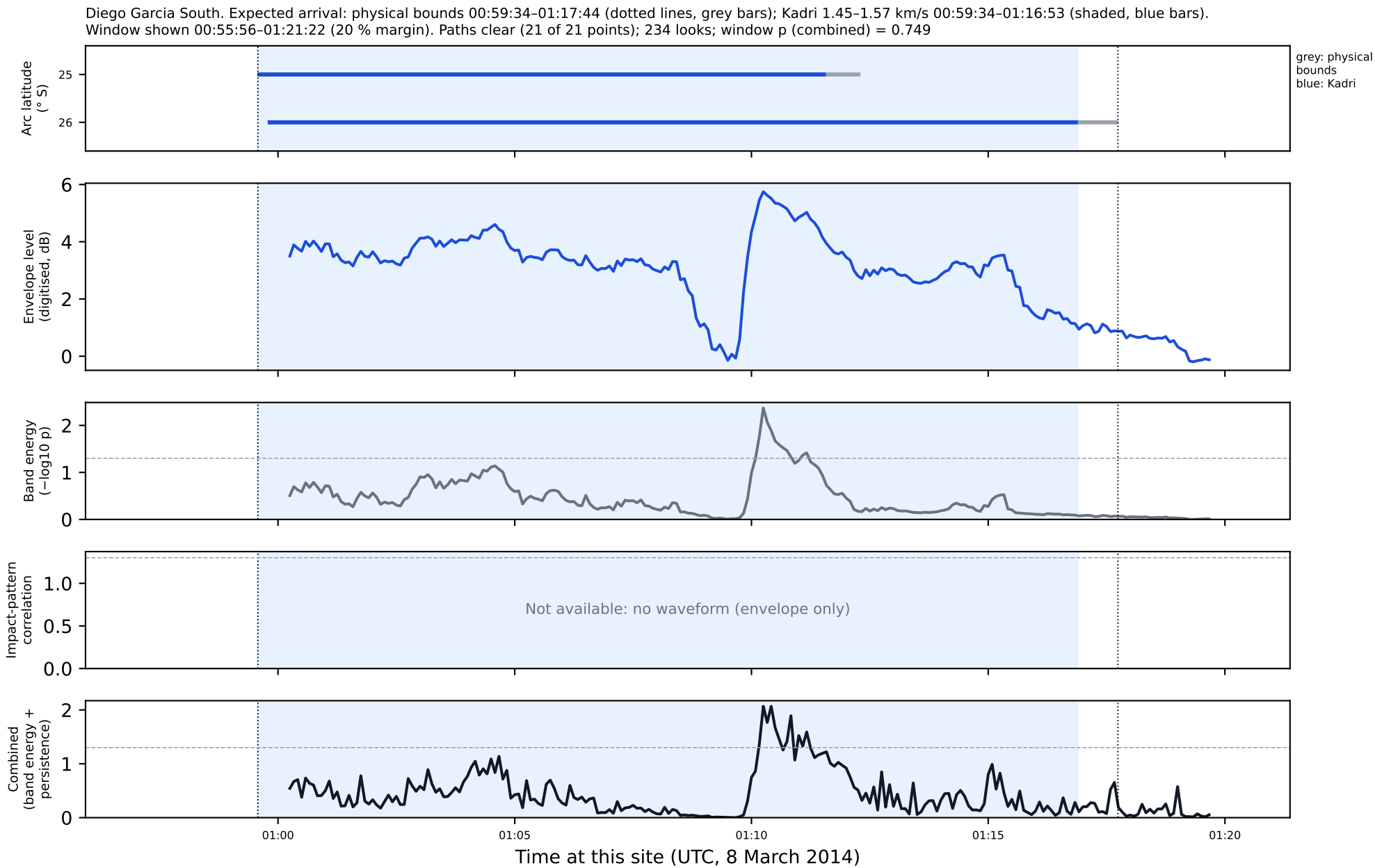
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Candidate 8: Cape Leeuwin 00:45:36 (Kadri; bearing 306.1° corrected, 312.0° published). Implied impact 00:23:02–00:31:40 UTC, arc 25; 26° S



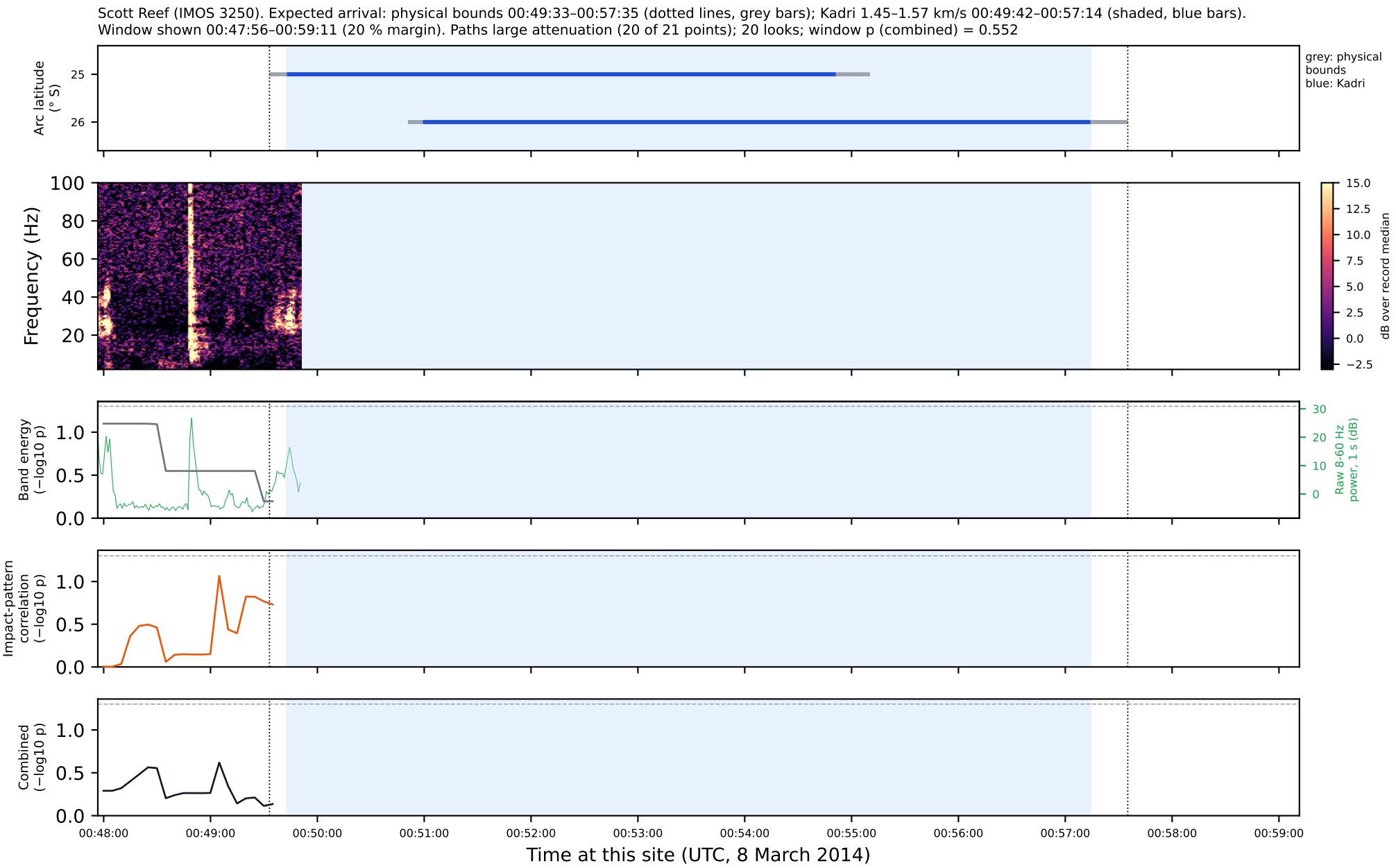
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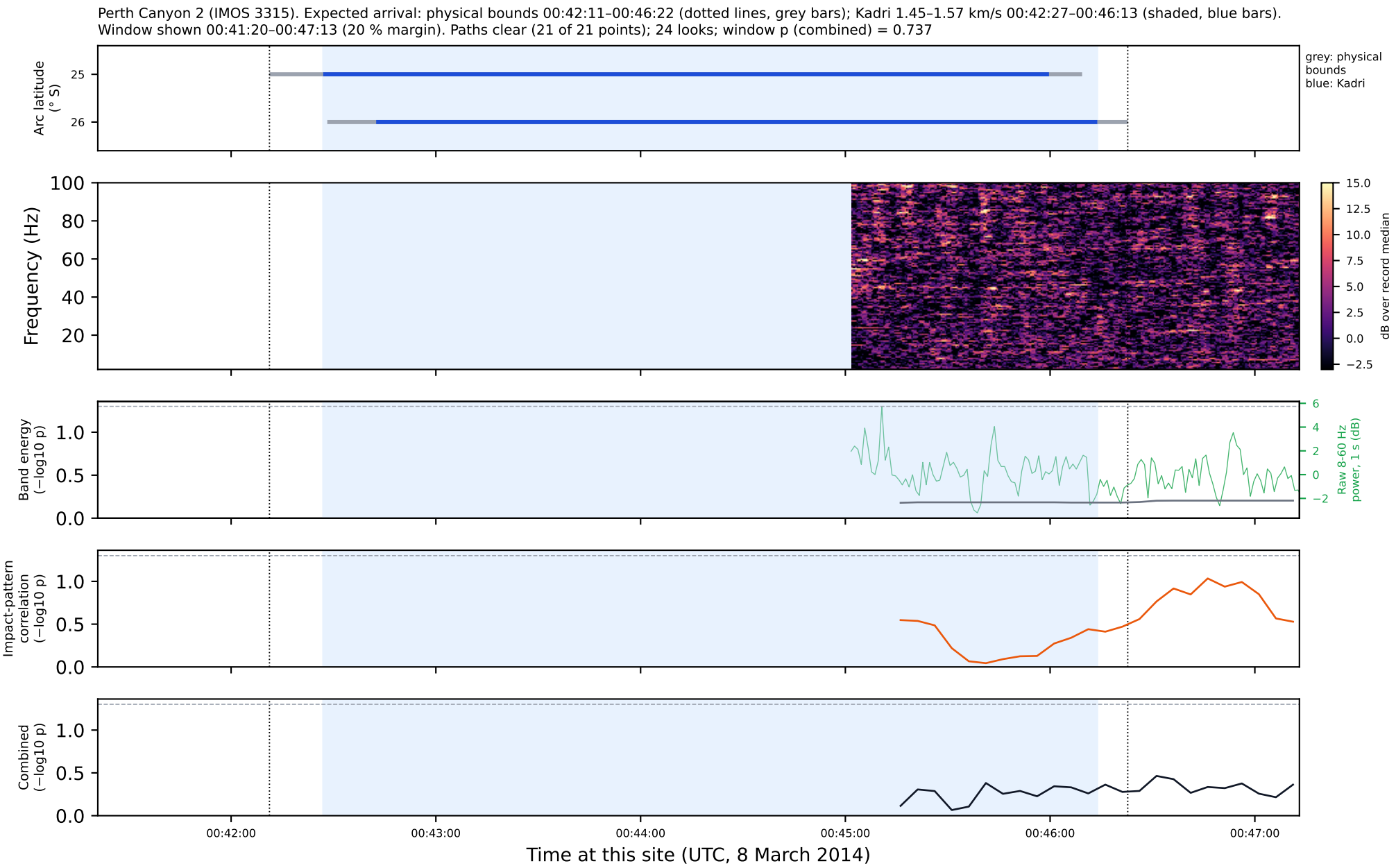
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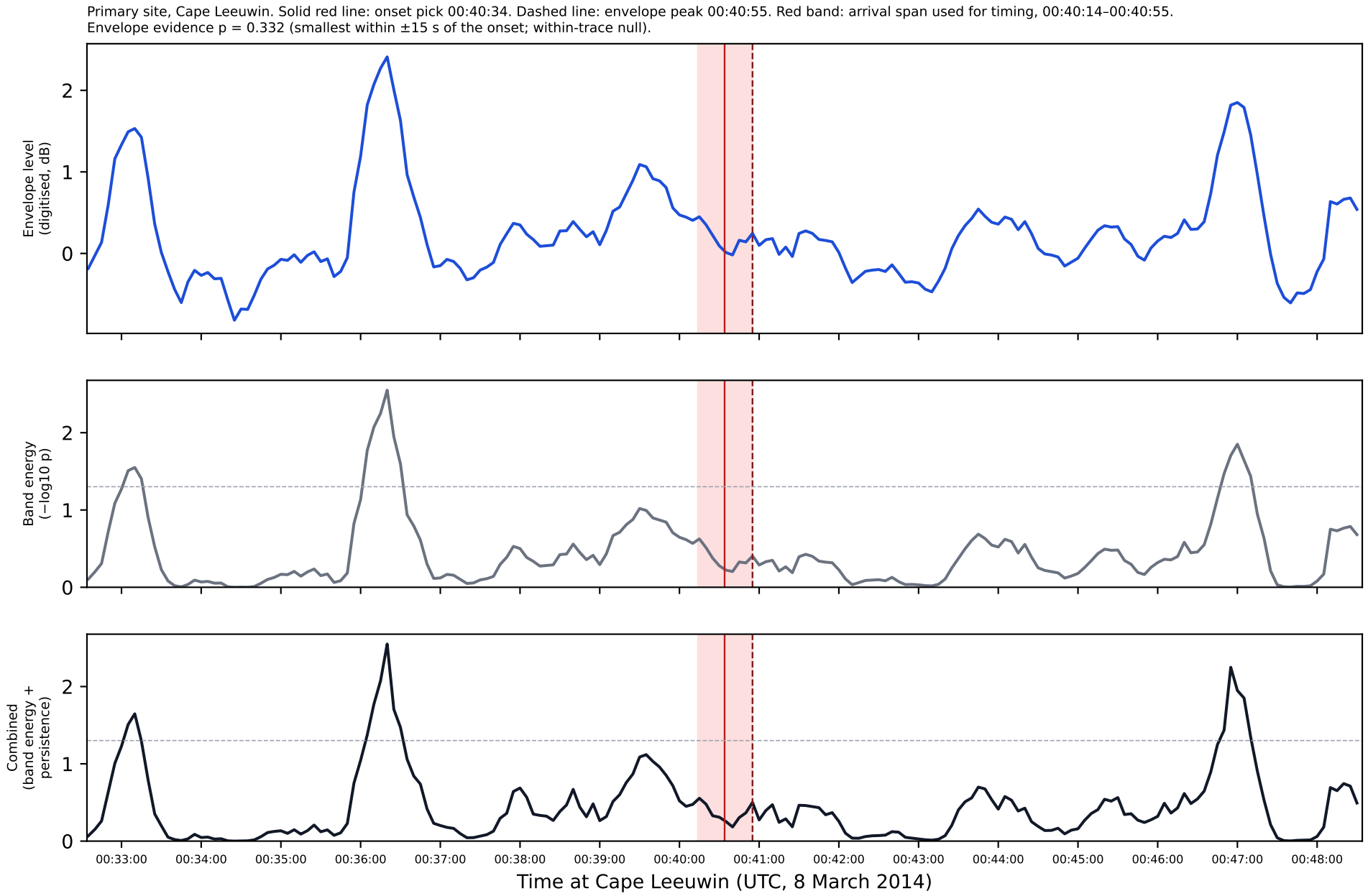
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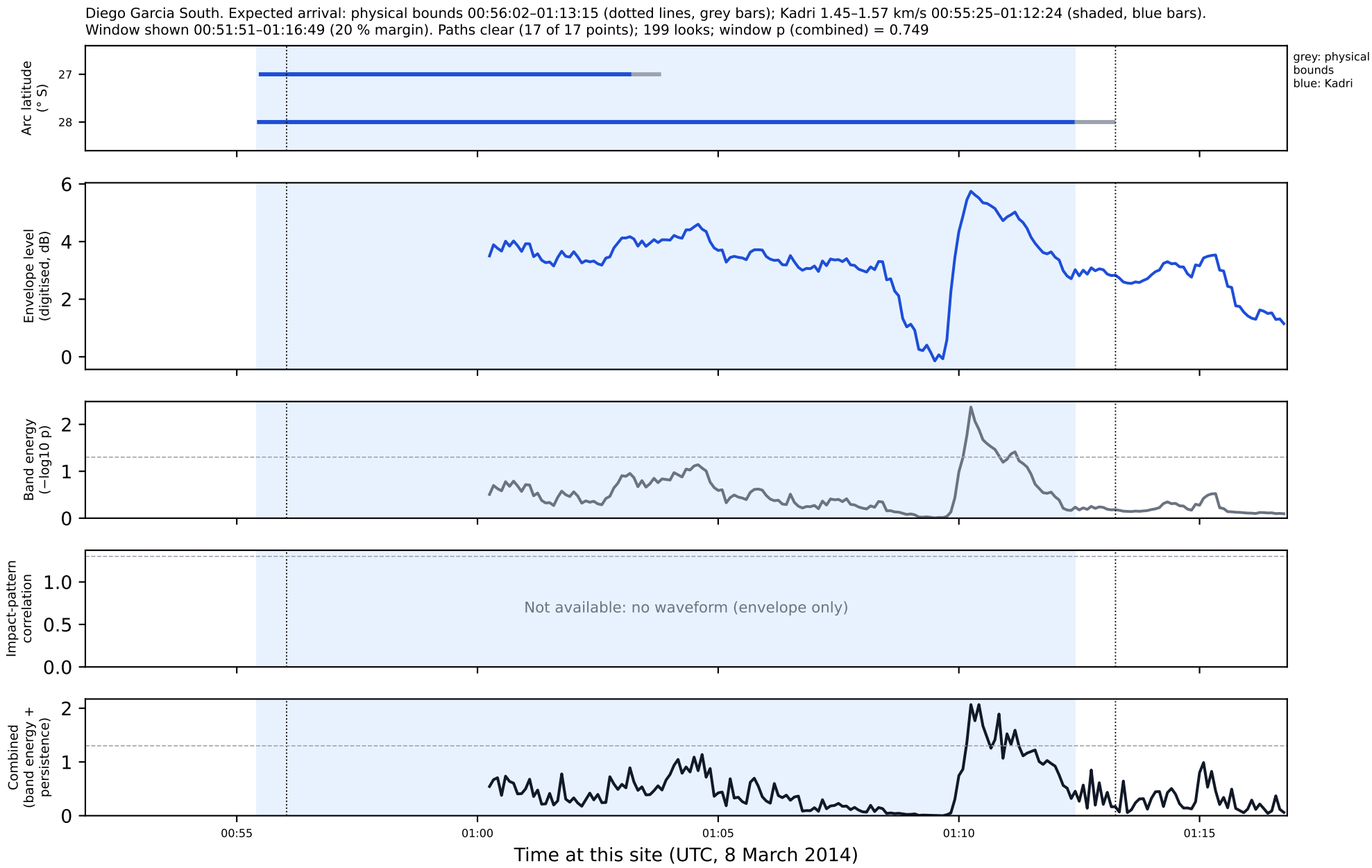
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Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:19:38–00:27:16 UTC, arc 28° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

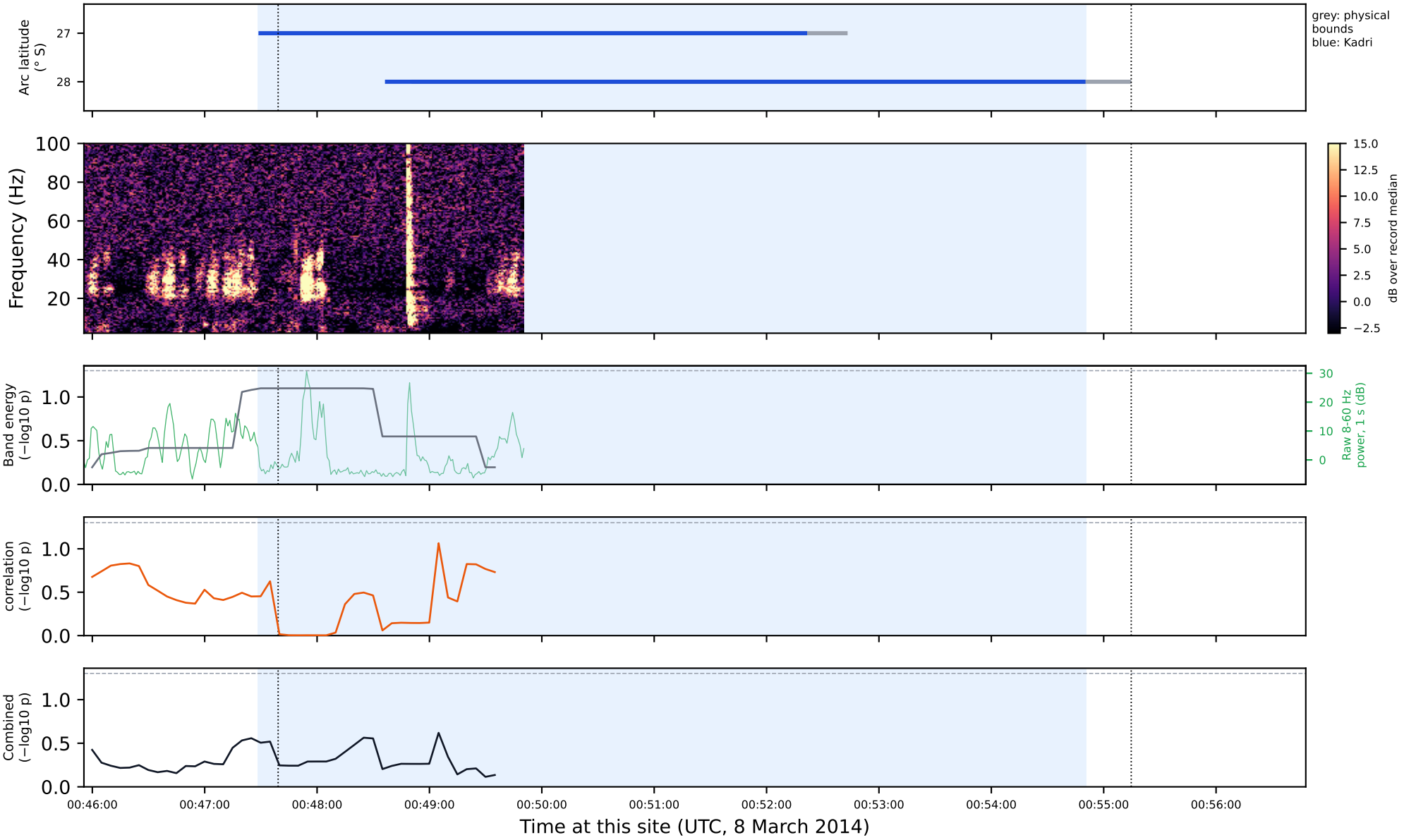
Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:19:38–00:27:16 UTC, arc 28° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

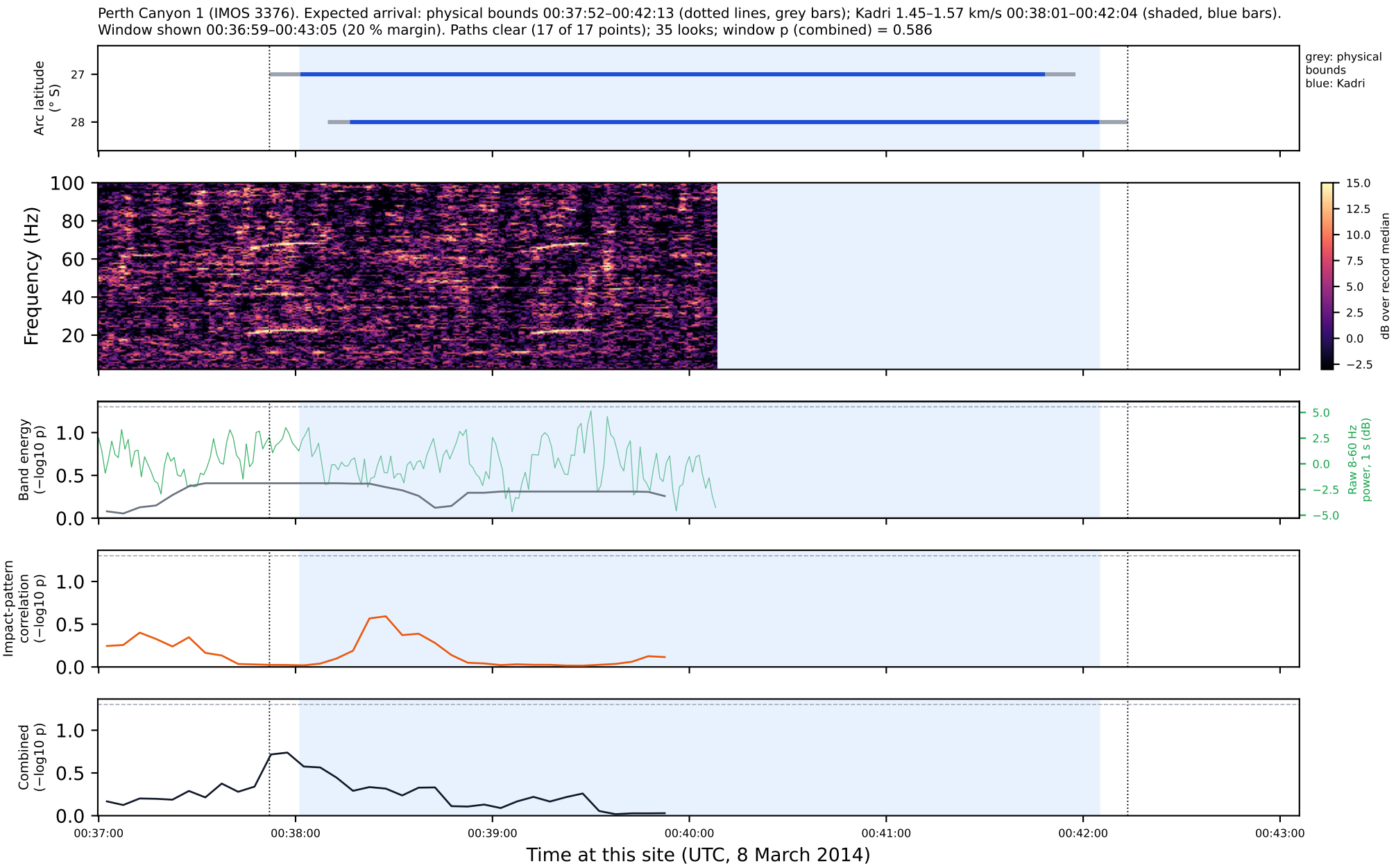
Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:19:38–00:27:16 UTC, arc 28° S

Scott Reef (IMOS 3250). Expected arrival: physical bounds 00:47:39–00:55:14 (dotted lines, grey bars); Kadri 1.45–1.57 km/s 00:47:28–00:54:50 (shaded, blue bars).
Window shown 00:45:55–00:56:47 (20 % margin). Paths large attenuation (13 of 17 points); 44 looks; window p (combined) = 0.752



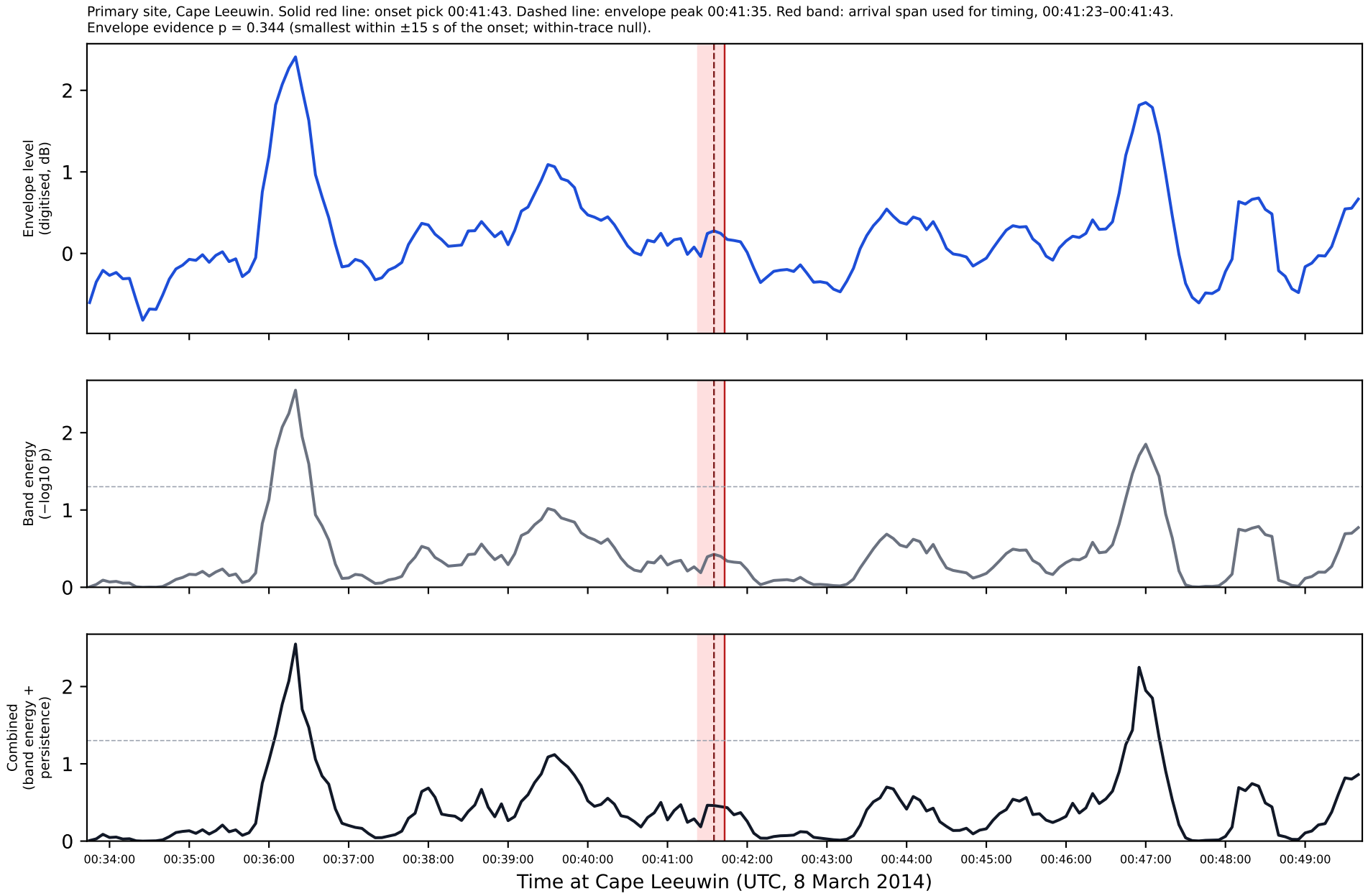
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:19:38–00:27:16 UTC, arc 28° S



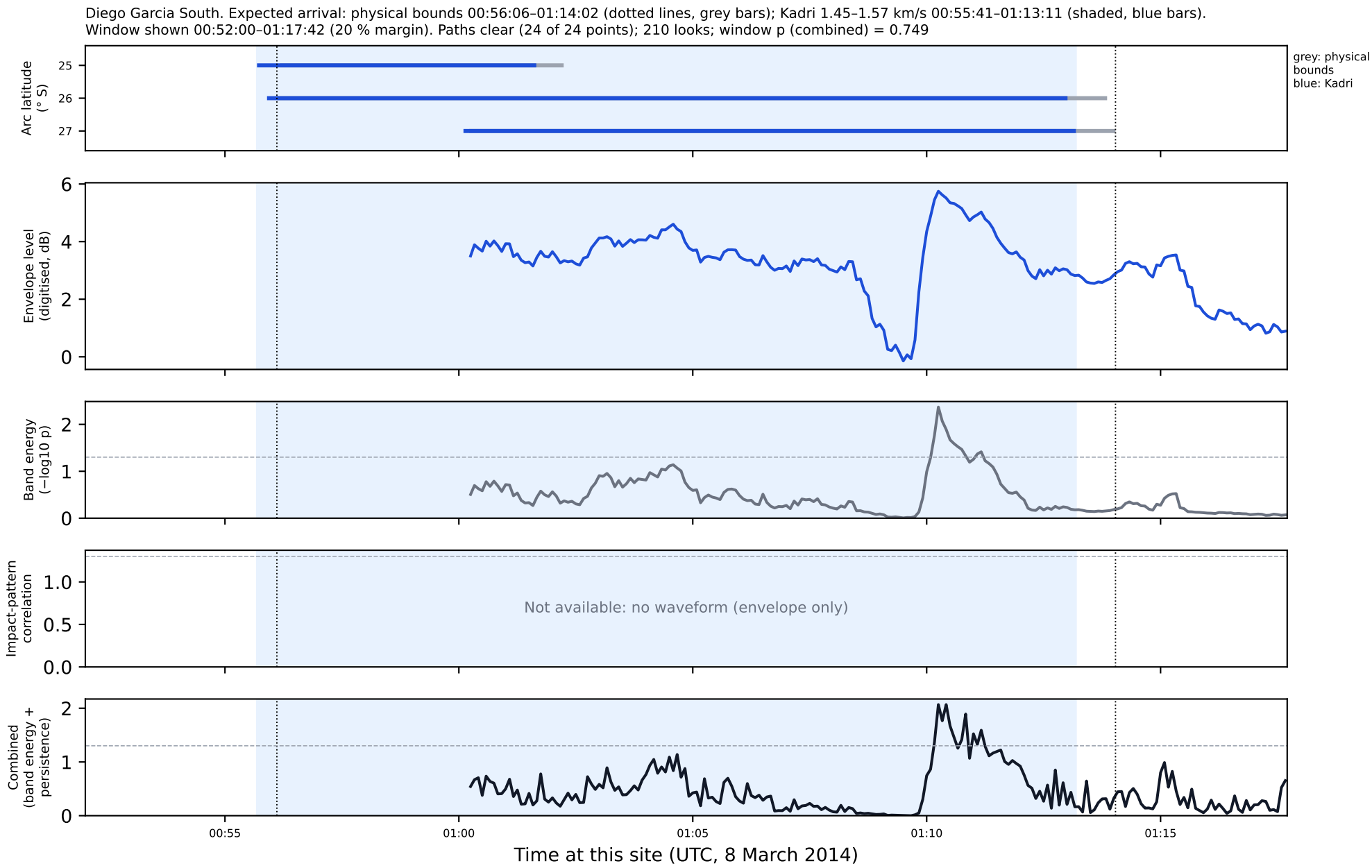
Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:19:38–00:28:01 UTC, arc 26; 27° S



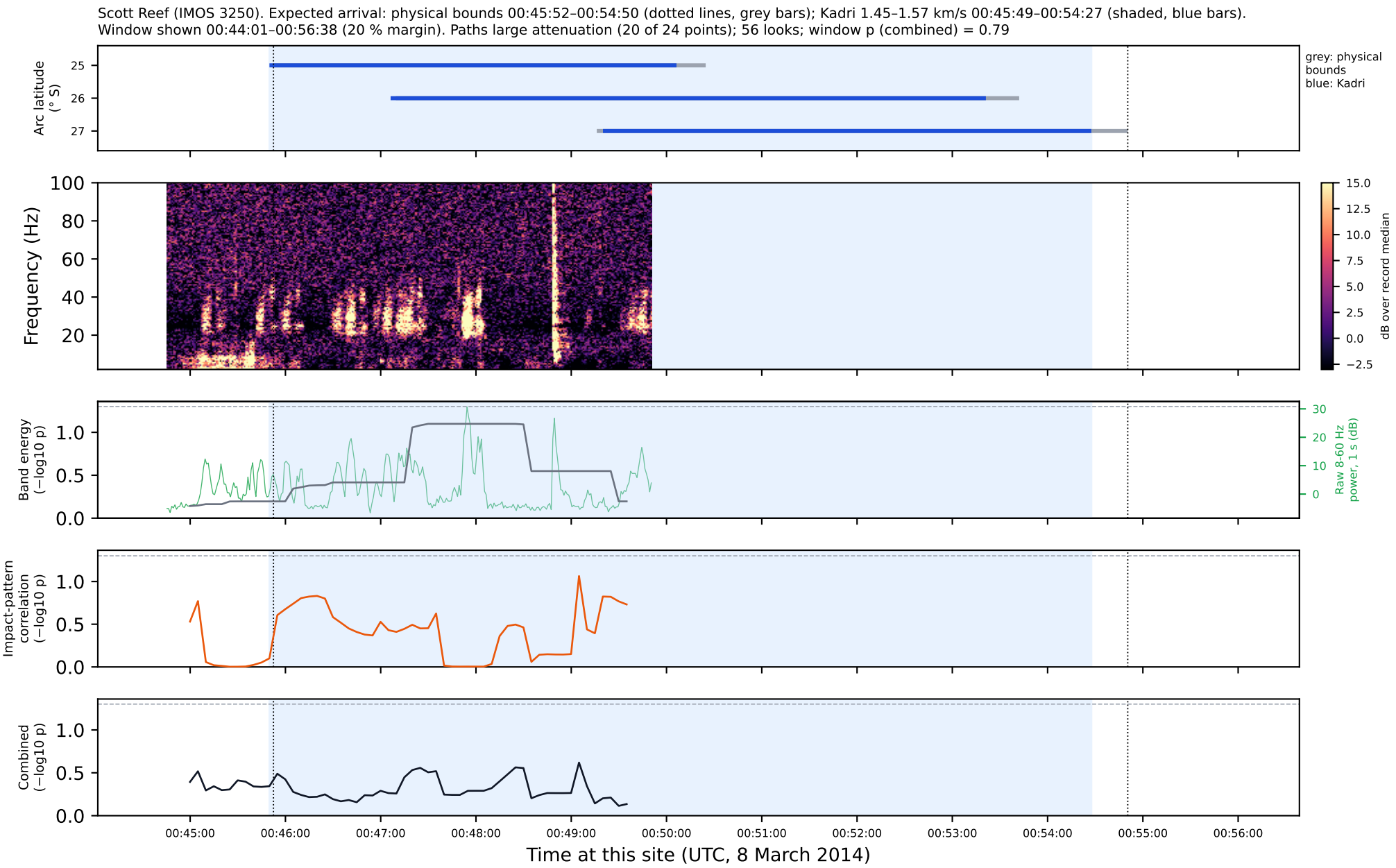
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset pick minus 20 s (Royer (2016) Cape Leeuwin figure pick bars sit 7.7-16.6 s after his Table 5 picks, up to 23.4 s for one array element; our envelope onsets match his figure bars for events A and B within 1.5 s) to the envelope peak; implied impact times use the start of the span with the slowest speed and the end of the span with the fastest speed. The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:19:38–00:28:01 UTC, arc 26; 27° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

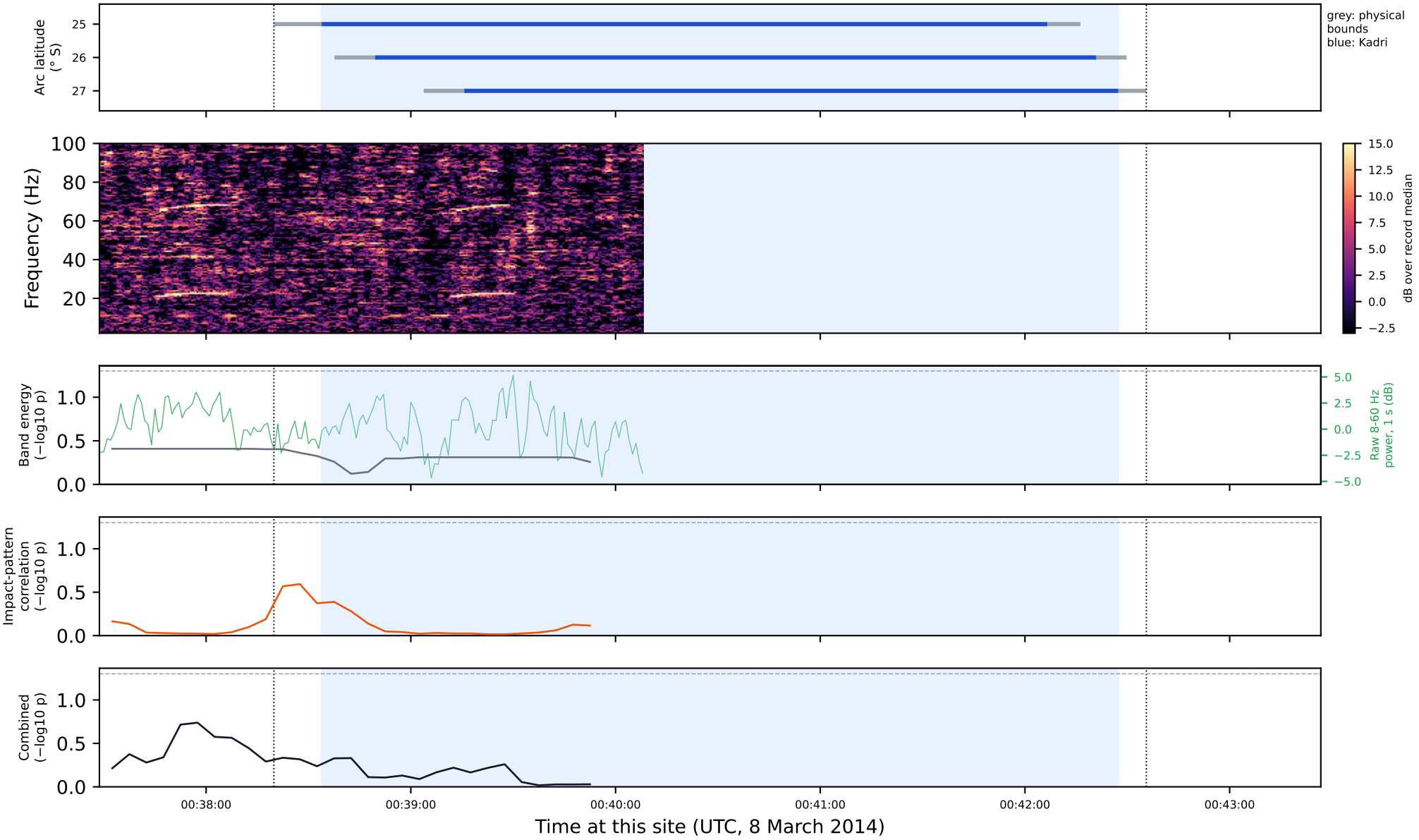
Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:19:38–00:28:01 UTC, arc 26; 27° S



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:19:38–00:28:01 UTC, arc 26; 27° S

Perth Canyon 1 (IMOS 3376). Expected arrival: physical bounds 00:38:19–00:42:35 (dotted lines, grey bars); Kadri 1.45–1.57 km/s 00:38:33–00:42:27 (shaded, blue bars).
Window shown 00:37:28–00:43:26 (20 % margin). Paths clear (24 of 24 points); 29 looks; window p (combined) = 0.536



Speeds. Blue: Kadri (2024) window 1.45 to 1.57 km/s, as published. Grey and dotted lines: physical bounds for sound carried in the water on each path. Slowest = path mean of the sound-channel axis speed minus 2 standard deviations, then 5 % longer path for detours; fastest = path mean of the fastest speed in the water column (surface or seabed) plus 2 standard deviations. Sound speed: WOA23 March climatology 2005-2014, 1 degree, sampled every 25 km on the great circle. Above about 1.55 km/s only a leg through rock could explain an arrival. Both spans cover every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing; arrival span from the onset minus 20 s to the envelope peak). Window shown: both spans together, widened by 20 % at each end. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).